

National Technical University of Athens

Department of Electrical and Computer Engineering

Modeling Smart Electric Vehicle Charging

**A Multi-Level Intelligence Approach with Renewable
Energy Integration**

Diploma Thesis

by

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Περίληψη

Η παρούσα διπλωματική εργασία αναλύει την έξυπνη φόρτιση ηλεκτρικών οχημάτων σε γραφειακούς χώρους που περιλαμβάνουν φωτοβολταϊκά συστήματα. Ο στόχος είναι να αξιολογηθεί η επίδραση διαφορετικών βαθμών ευφυΐας στο κόστος, στην εκμετάλλευση της τοπικής ανανεώσιμης ενέργειας και στην τήρηση των λειτουργικών περιορισμών του δικτύου. Η φόρτιση στον επαγγελματικό χώρο γίνεται ολοένα και πιο σημαντική, καθώς η αύξηση των ηλεκτρικών οχημάτων προκαλεί πίεση στα κτιριακά και δικτυακά συστήματα. Αυτό ισχύει ιδιαίτερα όταν η φόρτιση δεν συνοδεύεται από αυτοματισμούς. Τα κτίρια γραφείων είναι ιδανικοί χώροι εφαρμογής, καθώς προσφέρουν σταθερά ωράρια εργασίας, εκτεταμένα διαστήματα στάθμευσης και την ευκαιρία να αξιοποιηθούν πηγές ενέργειας από φωτοβολταϊκά συστήματα. Για αυτόν τον λόγο, δημιουργήθηκε ένα μοντέλο που ενσωματώνει το ηλεκτρικό φορτίο του κτιρίου, την παραγωγή ενέργειας από φωτοβολταϊκά συστήματα, την αποθήκευση ενέργειας, ένα στόλο ηλεκτρικών αυτοκινήτων και τους περιορισμούς του δικτύου. Η προσέγγιση στηρίζεται σε μια μελέτη περίπτωσης που χρησιμοποιεί ρεαλιστικά στοιχεία σχετικά με την κατανάλωση, την ηλιακή ακτινοβολία και τις τιμές του ηλεκτρικού ρεύματος. Μελετώνται μια βασισμένη σε κανόνες απλή στρατηγική και μια στρατηγική ενισχυτικής μάθησης χρησιμοποιώντας Q-learning. Επιπλέον, εξετάζεται η δυνατότητα επέκτασης σε λειτουργία Vehicle-to-Grid, που επιτρέπει στα οχήματα να επιστρέφουν ενέργεια στο κτίριο ή το δίκτυο. Η αξιολόγηση γίνεται με βάση οικονομικά, ενεργειακά, λειτουργικά και υπολογιστικά κριτήρια.

Τα αποτελέσματα αποκαλύπτουν ότι η οργανωμένη φόρτιση ενισχύει την απόδοση του συστήματος σε σύγκριση με μια βασική ανελαστική στρατηγική. Η χρήση απλών κανόνων εξασφαλίζει αξιόπιστη λειτουργία με χαμηλή πολυπλοκότητα και πλήρη τήρηση των βασικών περιορισμών. Η μέθοδος Q-learning παρέχει επιπλέον βελτίωση στα οικονομικά αποτελέσματα και μια πιο αποδοτική διαχείριση ενέργειας από το δίκτυο, ωστόσο προϋποθέτει περισσότερη υπολογιστική ισχύ και μια εκτενή διαδικασία εκπαίδευσης. Η συμβολή των φωτοβολταϊκών είναι θετική, αφού μειώνει την εξάρτηση από το δημόσιο δίκτυο και προάγει τη χρήση ανανεώσιμων πηγών ενέργειας σε τοπικό επίπεδο. Ωστόσο, το συνολικό όφελος που προσφέρουν εξαρτάται από τις διαστάσεις του συστήματος σε σύγκριση με τις ανάγκες φόρτισης. Η μελέτη του Vehicle-to-Grid αποκαλύπτει ότι η τεχνική ικανότητα δεν πάντα οδηγεί σε οικονομικό όφελος υπό όλες τις συνθήκες της αγοράς. Όταν η διαφορά ανάμεσα στις τιμές αγοράς και πώλησης είναι μικρή, τότε η διαδικασία αποφόρτισης γίνεται λιγότερο ελκυστική και τα οφέλη από το V2G παραμένουν περιορισμένα. Συνολικά, η εργασία καταλήγει

στο συμπέρασμα ότι η έξυπνη φόρτιση με τη χρήση φωτοβολταϊκών είναι μια ρεαλιστική και αποδοτική λύση για τα κτίρια γραφείων. Επιπλέον, οι πιο εξελιγμένες στρατηγικές ελέγχου και το V2G αποκτούν σημασία κυρίως όταν υποστηρίζονται από κατάλληλες πολιτικές και συνθήκες λειτουργίας.

Λέξεις κλειδιά: Ηλεκτρικά οχήματα, έξυπνη φόρτιση, ανανεώσιμες πηγές ενέργειας, βελτιστοποίηση, Vehicle-to-Grid, γραφειακά κτίρια.

Abstract

The research presented in this thesis examines the intelligent charging of electric vehicles at workplaces that have incorporated photovoltaic power plants into their design. The objective is to investigate how varying levels of "intelligence" influence both the cost associated with charging and the use of locally generated renewable energy as well as the degree to which the building's electrical loads conform to the grid operating requirements. With regard to workplace charging, there has been increasing interest in recent years due to the rapid proliferation of electric vehicles and its impact on both the infrastructure of individual buildings and grid operations when charging takes place in an uncontrolled manner. The reason why office buildings represent such an ideal area of investigation is due to the fact that these locations combine three factors that are ideally suited for the implementation of smart charging; namely, the predictability of work schedules, the relatively long duration of vehicle parking and the opportunity to incorporate energy generated through rooftop-mounted solar arrays.

In order to accomplish this, a mathematical model has been developed to simulate the behavior of an electric vehicle fleet that charges in a building equipped with energy storage, a PV system and subject to grid-related constraints. This model uses real-world data regarding energy usage and solar radiation as well as historical price data related to the purchase and sale of electricity. In addition, two types of control algorithms were evaluated: one type is a straightforward, rule-based algorithm and the other utilizes Q-learning as a means of developing an optimal control strategy. Finally, the ability of the vehicles to act as distributed generators and return excess energy to either the building or the grid (i.e., Vehicle-to-Grid operation) was investigated. From a financial perspective, this represents a potentially significant source of additional revenue that could help offset some portion of the total costs associated with owning and maintaining an electric vehicle. However, the extent to which Vehicle-to-Grid operation will provide a net benefit under various market conditions remains unclear.

For example, if the difference in price between purchasing electricity and selling electricity back to the utility company is not sufficient to cover the incremental cost associated with the development of V2G capabilities, then it may prove difficult to justify the expense of implementing such technology. Therefore, the remainder of this chapter evaluates each of these aspects of smart charging using a combination of economic, energetic, operational and computational metrics.

The findings reported in this thesis clearly demonstrate that cooperative charging signifi-

cantly improves system-wide performance relative to a simple inelastic strategy. The results further show that a rule-based approach provides reliable operation with minimal complexity and full compliance with most constraints. Additionally, the results show that a Q-learning approach provides improvements over the rule-based approach with respect to both economic efficiency and efficient management of grid energy. However, a major drawback of this method is that it requires substantially greater amounts of computational resources than do rule-based approaches and must undergo a training phase prior to operation. Furthermore, the analyses of photovoltaic contribution demonstrated that photovoltaic contribution is beneficial since it allows for a reduction in reliance upon grid-supplied electricity and increased utilization of locally-generated renewable energy; however, whether or not overall benefits derived from PV contribution are substantial ultimately depend upon relative sizes of the PV system(s) installed versus total peak-charging demand.

Keywords: Electric vehicles, smart charging, photovoltaic systems, reinforcement learning, optimization, Vehicle-to-Grid, office buildings.

List of Abbreviations

ANN	Artificial Neural Network
DDPG	Deep Deterministic Policy Gradient
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
ELMAS	European Load Model for Aggregated Sectors
ESS	Energy Storage System
EV	Electric Vehicle
kW	Kilowatt
kWh	Kilowatt-hour
MADRL	Multi-Agent Deep Reinforcement Learning
MPC	Model Predictive Control
PERC	Passivated Emitter and Rear Cell
PV	Photovoltaic
PVGIS	Photovoltaic Geographical Information System
RES	Renewable Energy Source
RL	Reinforcement Learning
SARAH3	Surface Solar Radiation Data Set (PVGIS database)
SoC	State of Charge
STC	Standard Test Conditions
V2G	Vehicle-to-Grid
Wp	Watt-peak (rated module power)

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Chapter 1

Introduction

1.1 Background and Motivation

Recent climate regulations have led to a rise in the adoption of electric vehicles. More and more people are opting for an electric vehicle for their daily commute. According to studies, most of the time spent in commute is on the way to and from the office. Also, most people spend a third of their time in offices, and with them their cars, which can be charged during the workday. The car batteries can also be used in times of peak consumption to give electricity back to the building or to the grid. Since the charging will happen throughout the day, it is logical to create smart charging algorithms that can utilise information from solar power generation, pricing spreads throughout the day, and grid capacity limits in order to optimise the charging schedule. Office buildings also offer a huge rooftop space for the installation of solar panels; the energy produced by these panels can be used both for the building's consumption as well as for the charging of the electric vehicles. The cost of electricity produced is zero, and this helps reduce the overall cost. As such, the utilization of renewable energy is the most important aspect of the charging algorithm.

1.2 Literature Review

There are many algorithms in play for the smart scheduling of the EV fleet. Early papers, such as Richardson [1], used linear programming to decide which vehicle will be charged when in 15-minute intervals, subject to network limitations. Later work included the use of reinforcement learning to improve grid stability, and optimise for pricing and renewables utilisation; several studies can be found in the work by Qiu et al. [2]. Below we discuss 10 representative papers from that review, all of which apply reinforcement learning to EV charging with the objective of cost minimisation and/or renewable energy integration, and compare to our present work. **Wan et al. (2019)** [3] proposed a model-free online drl-based real-time charging scheduling method for EV, using a representation network to extract features related to different time-varying electricity rates, as well as a Q-network that learns an action at every time step. They deal well with price uncertainty, but have a single EV and no renewable en-

ergy sources or building integration; we add a 15-EV fleet at an office building with rooftop PV and a stationary battery, and solve this problem using the same mechanism as the rest of this paper. **Shin et al. (2019)** [4] presented a multi-agent deep reinforcement learning for cooperative management of photovoltaic/energy storage system-enabled charging station. They do consider PV and energy storage, but we are more focused on intra-building optimisation than interstation. We model building consumption, grid capacity as an optimisation constraint, and V2G capability in a single integrated scenario. **Chiş et al. (2017)** [5] developed one of the earliest RL approaches for plug-in EV charging with forecasted price signals, using Q-learning to reduce charging costs. Their study considers a single EV and focuses on price optimisation; we include utility from solar energy as a further optimisation objective and handle a multi-vehicle fleet under a shared grid capacity constraint. **Qiu et al. (2020)** [6] propose a Prioritized DDPG method for the pricing of EVs with discrete charging levels, focusing primarily on the aggregator’s perspective. Their work captures the V2G flexibility and pricing economics, but does not model on-site renewable generation or building energy consumption. Our thesis instead approaches the problem from a building operator’s point of view, explicitly integrating renewable generation from rooftop PV production, and building loads into the reward signal. **Dorokhova et al. (2021)** [7] use deep reinforcement learning to control EV charging in a context with PV generation, comparing against rule-based and MPC benchmarks. Their formulation is the closest to ours in the list having PV integrated, but considers the problem of a single EV charging at a residential site. Our work differs in that we target a commercial office building with a fleet of 15 EVs, as well as a stationary battery, wholesale market electricity prices, and V2G capability. **Tuchnitz et al. (2021)** [8] suggest a smart charging strategy for an EV fleet using reinforcement learning to optimise fleet-level schedules and reduce the variance in the load sent to the grid by 65%. Their study focuses on the load flattening and resulting stability of the grid, but does not treat the problem of renewable self-consumption or cost optimisation; our work jointly optimises for both solar utilisation and cost, whilst also enforcing constraints on the grid. **Lee and Choi (2021)** [9] introduce a privacy-preserving deep reinforcement learning framework for dynamic pricing and energy management of multiple distributed charging stations with PV and ESS. Their work addresses the pricing problem from the station operator’s perspective using federated learning techniques. Our work differs in that we optimise from the point of view of the building operator, and use tabular Q-learning (which is understandable), rather than DDPG. We evaluate the profitability of V2G under real European wholesale prices. **Abdullah et al. (2021)** [10] offer an overview of RL-based EV charging management systems, categorising the algorithms used and comparing objectives. Their survey highlights that most studies opt for only one objective (cost, PV output) when in practice a complete algorithm that considers all is of much greater utility, although this kind of study is also hard to do under realistic assumptions. Our thesis helps fill in this space by performing a complete, integrated simulation on real European data, using a two-level approach and evaluating V2G viability. **Sadeghi-**

anpourhamami et al. (2018) [11] have introduced an scalable model-free demand response strategy using batch reinforcement learning (Fitted Q-Iteration) for coordinated charging of fleets. Due to its compact state representation, the scalability of Sadeghianpourhamami et al.'s (2018) approach is supported, however the focus of the reward function lies entirely on the load smoothing. In contrast to this, we have formulated the Q-learning to take into account a multi-objective reward with costs, solar usage, target SoC achievement and penalty terms for violations of constraints.

Park and Moon (2022) [12] presented a multi-agent DRL strategy based on centralized training and decentralized implementation for scheduling EV charging. Their method allows multi-vehicle coordination via attention networks. However, they do not take into account on-site renewable generation nor V2G. We show here that even a much simpler tabular Q-learning method can achieve significant cost savings (of 3.7%) while achieving the same level of target SoC as Park and Moon (2022), if the building PV generation and dynamic pricing data are incorporated appropriately into the Q-learning.

1.2.1 Summary and Contribution of This Thesis

The literature review summary of the reviewed papers including a comparison to the current thesis is given in Table 1.1.

Table 1.1: Comparison of reviewed works with the present thesis

Study	RL Method	Price Opt.	PV/RES	V2G	Fleet	Building	Battery
Wan et al. (2019)	DQN	✓					
Shin et al. (2019)	MADRL	✓	✓				✓
Chiş et al. (2017)	Q-learning	✓					
Qiu et al. (2020)	DDPG	✓		✓			
Dorokhova et al. (2021)	DRL		✓				
Tuchnitz et al. (2021)	RL+ANN				✓		
Lee & Choi (2021)	Fed-DRL	✓	✓				✓
Abdullah et al. (2021)	Survey	—	—	—	—	—	—
Sadeghianpourhamami et al. (2018)	Batch RL				✓		
Park & Moon (2022)	MADRL	✓			✓		
This thesis	Q-learning	✓	✓	✓	✓	✓	✓

As indicated in Table 1.1, the present work is distinctive as follows:

- (i) A combination of algorithms with various complexity levels from simple rules to Q-learning and V2G-enabled Q-learning.
- (ii) Integration of on-site rooftop PV generation into the reward function.
- (iii) Stationary building battery.
- (iv) Realistic mobility patterns for a 15-EV fleet.
- (v) Real European wholesale electricity prices.
- (vi) Explicit modeling of office building consumption.

Most previous studies addressed at most three of these characteristics.

1.3 Smart Charging Levels

The algorithms developed in this thesis are separated by their intelligence levels. Level 0 is a simple rule-based algorithm that prioritises solar renewable energy, respecting grid constraints and the target charging SOC of the EV battery; it charges without respecting future price levels. Level 1 is a Q-learning algorithm that uses a reward function to include information about price levels and also the availability of solar energy. Level 2 is a Q-learning algorithm that also includes the action of discharging energy back to the grid. The remaining chapters focus on different parts of the algorithms.

1.4 Thesis Structure

Chapter 2 focuses more on the models that describe the office, the EVs, the grid, and the solar panels. Chapter 3 focuses on the data for the office building consumption, electricity pricing, and solar generation. Chapter 4 focuses on the simulation using each algorithm, collecting and comparing the results. Chapter 5 focuses on the conclusion of the results and recommendations for future work.

Chapter 2

Methodology

This chapter describes the methodology used to simulate a real environment. It includes the models and equations used to describe the different components. The different algorithms are also described in this chapter in order of intelligence levels.

2.1 System Components

2.1.1 Office Building Model

The office building is modeled first by its energy consumption. The stationary battery of the building is also represented by a battery model. The solar panels are represented by a solar panel model, and the grid is represented by a grid model.

Energy Consumption

For the building's energy consumption $C_b(t)$, we utilize the ELMAS dataset (European Load Model for Aggregated Sectors), specifically the Office cluster (#5). Unfortunately, the dataset only contains mean data for the classes of the cluster, so we calculated a sample building's annual energy. We then extracted this data into the `Time_series_18_clusters.csv` file. The following processing steps were followed:

Let L_t be the mean hourly energy load of the different classes of the Office cluster ($T = 8,736$ hours for the year 2018) with yearly energy:

$$E_{cluster} = \sum_{t=1}^T L_t \quad (2.1)$$

To be able to represent a single office building, we calculate the per-customer annual energy for each class c belonging to the Office cluster column:

$$e_c = \frac{E_c}{N_c} \quad (2.2)$$

where E_c is the annual energy and N_c the number of customers in class c . The 75th

percentile is used to scale the energy to represent a medium to big office.

$$E_{target} = Q_{0.75}(\{e_c\}) = 58,831 \text{ kWh} \quad (2.3)$$

The energy profile L_t is then scaled using the energy target of the single building:

$$L_t^{scaled} = L_t \cdot \frac{E_{target}}{E_{cluster}} \quad (2.4)$$

To simulate a more realistic scenario, we also introduce noise to the energy profile. We use a daily factor α_d (constant within each day) and an hourly factor β_t :

$$\alpha_d \sim \mathcal{N}(1, \sigma_d^2), \quad \sigma_d = 0.08, \quad \alpha_d \in [0.7, 1.3] \quad (2.5)$$

$$\beta_t \sim \mathcal{N}(1, \sigma_h^2), \quad \sigma_h = 0.03, \quad \beta_t \in [0.9, 1.1] \quad (2.6)$$

The synthetic series before re-normalization:

$$L_t^{syn} = L_t^{scaled} \cdot \alpha_d(t) \cdot \beta_t \quad (2.7)$$

Finally, as we did in the previous step, we re-scale the series to preserve the target annual energy:

$$C_b(t) = L_t^{syn} \cdot \frac{E_{target}}{\sum_{t=1}^T L_t^{syn}}. \quad (2.8)$$

The single-office profile has a mean load of 6.73 kW and a peak load of 11.92 kW.

Solar Photovoltaic System

The solar PV system is modeled by the following equation:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t), \quad (2.9)$$

where:

- A_{panel} = solar panel area (m²)
- η_{panel} = panel efficiency
- $I(t)$ = solar irradiance at hour t (W/m²)

Building Battery Storage

The building's battery is modeled as follows:

$$P_{dis,t} = \min(P_{max}, (\text{SoC}_{t-1} - \text{SoC}_{min}) \cdot \eta) \quad (2.10)$$

$$P_{ch,t} = \min(P_{max}, C - \text{SoC}_{t-1}) \quad (2.11)$$

$$\text{SoC}_t = \text{SoC}_{t-1} + \min(P_{ch,t}, E_{excess}(t)) \quad (2.12)$$

$$\text{SoC}_t = \text{SoC}_{t-1} - \min\left(\frac{P_{dis,t}}{\eta}, \frac{E_{missing}(t)}{\eta}\right) \quad (2.13)$$

where:

- P_{max} = maximum charge/discharge power [kW]
- C = battery capacity [kWh]
- SoC_{min} = minimum allowable state of charge
- η = round-trip efficiency
- $E_{excess}(t) = \max(0, P_{solar}(t) - C_b(t))$ = surplus solar energy
- $E_{missing}(t) = \max(0, C_b(t) - P_{solar}(t))$ = energy deficit

Constraints:

$$\text{SoC}_{min} \leq \text{SoC}_t \leq C \quad (2.14)$$

$$0 \leq P_{ch,t} \leq P_{max} \quad (2.15)$$

$$0 \leq P_{dis,t} \leq P_{max} \quad (2.16)$$

Net Energy Demand

The building's net energy demand (positive = needs grid, negative = surplus):

$$E_{net}^{building}(t) = C_b(t) - P_{solar}(t) - E_{discharge}^b(t) + E_{charge}^b(t). \quad (2.17)$$

2.1.2 Electric Vehicle Fleet Model

Individual EV Characteristics

Each EV i in the fleet has the following information:

- Battery capacity: B_{cap}^i [kWh]

- State of charge: $\text{SoC}^i(t)$ [dimensionless, 0–1]
- Maximum charge rate: $P_{max,charge}^i$ [kW]
- Maximum discharge rate: $P_{max,discharge}^i$ [kW] (for V2G)
- Arrival time: $t_{arrival}^i$ [hour]
- Departure time: t_{target}^i [hour]
- Desired SoC: $\text{SoC}_{desired}^i$ [dimensionless]

EV Battery Dynamics

The state of charge evolves for EV i following the equation:

$$\text{SoC}^i(t+1) = \begin{cases} \text{SoC}^i(t) + \frac{\eta_{charge} E_{charge}^i(t)}{B_{cap}^i} & \text{when charging } (E_{discharge}^i(t) = 0), \\ \text{SoC}^i(t) - \frac{E_{discharge}^i(t)}{\eta_{discharge} B_{cap}^i} & \text{when discharging } (E_{charge}^i(t) = 0). \end{cases} \quad (2.18)$$

where η_{charge} and $\eta_{discharge}$ are charging and discharging efficiencies (typically 0.95).

Constraints

Each EV must satisfy:

$$\text{SoC}_{min} \leq \text{SoC}^i(t) \leq 1.0 \quad \forall t \quad (2.19)$$

$$\text{SoC}^i(t_{target}^i) \geq \text{SoC}_{desired}^i \quad (2.20)$$

$$E_{charge}^i(t) = 0 \quad \text{if } t < t_{arrival}^i \text{ or } t \geq t_{target}^i \quad (2.21)$$

Constraint (2.20) ensures that each EV reaches its desired charge level before departure.

2.1.3 Grid Model

Dynamic Pricing

For the simulations, we use dynamic pricing with a different price for each hour:

$$\pi_{buy}(t) = \text{price_profile}[t \bmod 24] \quad [\text{€/kWh}]. \quad (2.22)$$

For V2G scenarios, the selling prices are 80% of the buying prices:

$$\pi_{sell}(t) = \text{sell_price_profile}[t \bmod 24] \quad [\text{€/kWh}]. \quad (2.23)$$

Typically, $\pi_{sell}(t) \leq \pi_{buy}(t)$ (80-90% of purchase price).

Grid Capacity Constraint

The grid provides a certain capacity that can be used by the building for charging:

$$E_{grid}^{building}(t) + \sum_{i=1}^{N_{EV}} E_{grid}^i(t) \leq P_{grid,max}(t), \quad (2.24)$$

where $P_{grid,max}(t)$ represents the hourly grid capacity limit.

2.2 Optimization Problem Formulation

2.2.1 Objective Function

The goal is to minimize the total electricity cost:

$$\min \sum_{t=0}^{T-1} [\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot E_{V2G}^{total}(t)], \quad (2.25)$$

where:

- $E_{grid}^{total}(t)$ = total energy purchased from grid at hour t
- $E_{V2G}^{total}(t)$ = total energy sold back to grid (V2G only)
- T = simulation duration (24 hours)

2.2.2 Energy Balance

For each hour t , the energy equation must be satisfied:

$$C_b(t) + \sum_{i=1}^{N_{EV}} E_{charge}^i(t) = P_{solar}(t) + E_{grid}^{total}(t) + \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \quad (2.26)$$

2.3 Level 0: Baseline (Uncontrolled Charging)

2.3.1 Algorithm Description

The simple algorithm prioritizes the remaining solar energy and respects grid capacity; each car charges at the maximum rate upon arrival until its desired state of charge is reached:

Algorithm 1 Simple Rule-Based Charging

```

1: for each hour  $t$  do
2:   Calculate building surplus:  $E_{surplus}(t) = \max(0, P_{solar}(t) - C_b(t))$ 
3:   Calculate available grid capacity:  $P_{available}(t) = P_{grid,max}(t) - \max(0, C_b(t) - P_{solar}(t))$ 
4:    $E_{remaining} \leftarrow E_{surplus}(t)$ 
5:   for each EV  $i$  (in order) do
6:     if  $t_{arrival}^i \leq t < t_{target}^i$  and  $SoC^i(t) < SoC_{desired}^i$  then
7:        $E_{needed}^i = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i)$ 
8:        $E_{from\_surplus} = \min(E_{needed}^i, E_{remaining})$ 
9:        $E_{from\_grid} = \min(E_{needed}^i - E_{from\_surplus}, P_{available}(t))$ 
10:       $E_{total} = E_{from\_surplus} + E_{from\_grid}$ 
11:      Charge EV  $i$  with  $E_{total}$ 
12:      Update  $E_{remaining}$  and  $P_{available}(t)$ .
13:    end if
14:  end for
15: end for
    
```

2.3.2 Mathematical Formulation

Energy for EV i at hour t :

$$E_{charge}^i(t) = \min(P_{max,charge}^i, (SoC_{desired}^i - SoC^i(t)) \cdot B_{cap}^i, E_{available}(t)) \quad (2.27)$$

where $E_{available}(t)$ contains both the building's surplus energy and the remaining grid capacity.

2.4 Level 1: Intelligent Optimization Algorithms

2.4.1 Reinforcement Learning (Q-Learning)

State Space

The state for EV i at hour t is represented by the following vector:

$$s^i(t) = (EV_id, SoC_{bin}^i(t), t, grid_availability), \quad (2.28)$$

where $SoC_{bin}^i(t)$ is the discretized SoC (bins of 0.1).

Action Space

For charge-only systems:

$$\mathcal{A} = \{\text{charge, standby}\}. \quad (2.29)$$

For V2G-enabled systems:

$$\mathcal{A}_{V2G} = \{\text{charge, discharge, standby}\}. \quad (2.30)$$

Reward Function

The reward function is as follows:

$$\begin{aligned} R(s, a, s') = & -\alpha_1 \cdot E_{grid}^i(t) \cdot \pi_{buy}(t) \quad (\text{minimize grid cost}) \\ & + \alpha_2 \cdot E_{solar}^i(t) \quad (\text{maximize solar usage}) \\ & + \alpha_3 \cdot \mathbb{K}[\text{SoC}^i(t_{target}) \geq \text{SoC}_{desired}^i] \quad (\text{target achievement}) \\ & - \alpha_4 \cdot \max(0, \text{SoC}_{desired}^i - \text{SoC}^i(t_{target})) \quad (\text{penalty for shortfall}) \\ & - \alpha_5 \cdot \mathbb{K}[\text{SoC}^i(t) < \text{SoC}_{min}] \quad (\text{constraint violation}) \end{aligned} \quad (2.31)$$

where $\alpha_1, \dots, \alpha_5$ are weights.

Q-Learning Update Rule

The Q-value is updated using the temporal difference method:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (2.32)$$

where:

- α = learning rate (0.1-0.15)
- γ = discount factor (0.95-1.0)
- s' = next state after taking action a in state s

Policy Execution

After training, the policy selects actions greedily:

$$a^*(s) = \arg \max_{a \in \mathcal{A}} Q(s, a). \quad (2.33)$$

2.5 Level 2: Vehicle-to-Grid (V2G)

2.5.1 Bidirectional Energy Flow

V2G expands the optimization to include discharge decisions:

$$E_{net}^i(t) = E_{charge}^i(t) - E_{discharge}^i(t). \quad (2.34)$$

2.5.2 V2G Constraints

Additional constraints for V2G are as follows:

$$E_{discharge}^i(t) \leq (\text{SoC}^i(t) - \text{SoC}_{desired}^i) \cdot B_{cap}^i \quad (\text{preserve target SoC}) \quad (2.35)$$

$$E_{charge}^i(t) \cdot E_{discharge}^i(t) = 0 \quad (\text{no simultaneous charge/discharge}). \quad (2.36)$$

2.5.3 Revenue Calculation

Total benefit with V2G:

$$\text{Benefit}_{V2G} = - \sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot \sum_{i=1}^{N_{EV}} E_{discharge}^i(t) \right]. \quad (2.37)$$

Positive benefit indicates revenue or cost saving.

2.5.4 V2G Integration with Algorithms

The Q-Learning algorithm is extended to handle V2G by adding the action of discharging and by adding the selling reward to the reward function; additionally, discharge constraints are included.

Chapter 3

Case Study: Office Building Scenario

The case study models a single office building with solar panels installed in the rooftop, 15 electric vehicles charging at the building, a stationary battery storage system, and a grid connection with capacity limits. The simulation window is one day.

3.1 Building Energy Profile

3.1.1 Energy Consumption Data

Data Source

The energy consumption profile as stated in the previous chapter was derived from ELMAS dataset. A new timeseries was created with synthetic data.

Profile Characteristics

Figure 3.1 shows the hourly consumption pattern:

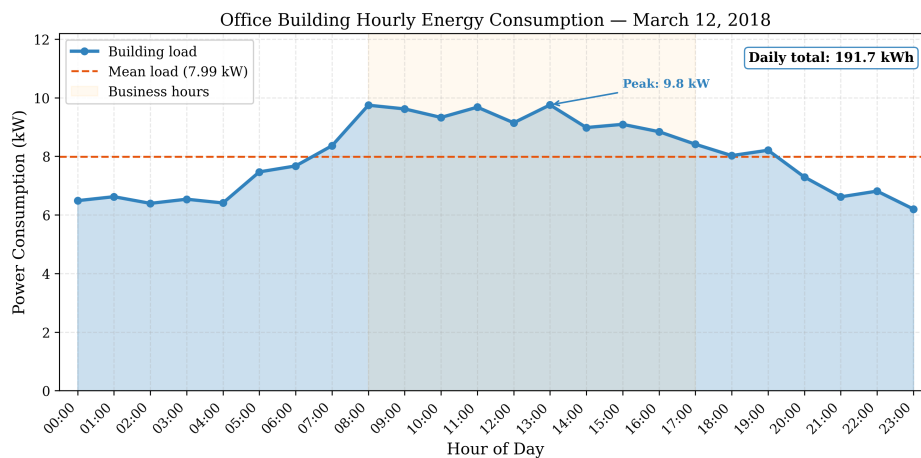


Figure 3.1: Office building hourly energy consumption profile

Annual statistics are a mean load of 6.73 kW, a peak load of 11.92 kW during business

hours, an annual energy of 58,831 kWh, and a daily consumption of about 161 kWh (annual divided by 365).

Validation

The consumption profile comes from the ELMAS source data. The per-customer annual energy distribution, with mean 10,429 kWh and $P75 = 58,831$ kWh, reflects small to medium-sized office buildings. The 75th percentile is used to represent a medium-to-large office. The cluster shape has a clear working-hours peak and a low night baseline, which matches an office profile, and the added noise keeps the curve from looking overly smooth while preserving the annual total.

3.1.2 Solar PV System

System Parameters

Values of the rooftop solar PV system are:

Table 3.1: Solar PV system specifications

Parameter	Value
Panel area	150 m ²
Panel efficiency	20%
Peak power	30 kW
Technology	Monocrystalline PERC (JA Solar DeepBlue 3.0 Pro)
Reference module	JAM54S31-405/MR (405 Wp, 20.7% efficiency)
No. of modules	74

The manufacturer data (for the reference module) is from JA Solar DeepBlue 3.0 Pro (JAM54S31-405/MR) [13] - a popular monocrystalline PERC half-cut cell panel type, rated at 405 Wp with a module efficiency of 20.7% (STC). We have installed 74 units, with a total approximate unit area of $144.5\text{m}^2 \approx 150\text{m}^2$ and still a peak system power of $29.97\text{kW} \approx 30\text{kW}$.

Irradiance data

The solar irradiance data is derived from PVGIS (Photovoltaic Geographical Information System) [14]. It is also the European Commission Joint Research Centre PVGIS database. Selected the location Paris, France (48.857°N, 2.352°E, 32 m elevation) and the PVGIS-SARAH3 radiation database. Module oriented to South (azimuth 0°) with a tilt of 37° (site-optimum). Selected March 12, 2018 as a representative Spring Day. Hourly data file is `irradiance_pvgis.csv`.

Solar Production profile

Hourly solar power production:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) = 150 \text{ m}^2 \cdot 0.20 \cdot I(t) \quad (3.1)$$

Profile is represented by:

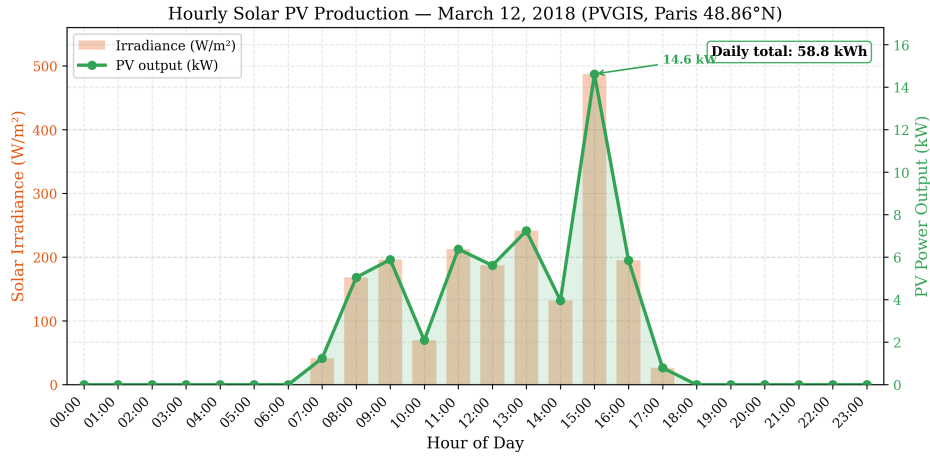


Figure 3.2: Hourly solar PV production profile (12th March 2018)

Maximum production at about 14.6 kW at 15:00 when maximum irradiance measured is 487 W/m²; Total daily production ≈58.8kWh. Part of the building consumption for business hours is offset (≈7:00–17:00) and appears to be surplus at 15:00 as well. Additionally, the profile compliments the EV parking hours (6 AM - 6PM) aspect of the profile. There is also regularly variable irradiance due to intermittent cloud cover typically observed in Paris in Spring.

Justification for Selected date

March 12, 2018 was selected because it is a Spring day with midrange curtailment of net demand and midrange clouds accounted in Paris PVGIS data. This rule also helps avoid using extreme Summer or Winter days inappropriately; It also helps minimise further methodology errors. SARA3 clouds for this date are well validated overall.

3.1.3 Building Battery Storage

The building contains a stationary battery storage system:

Table 3.2: Building battery specifications

Parameter	Value
Capacity	80 kWh
Efficiency	95%
Initial SoC	50%
Depth of discharge limit	80%
Max charge rate	50 kW
Max discharge rate	50 kW
Chemistry	Lithium Iron Phosphate (LFP)
Reference product	BYD Battery-Box Premium HVM

The battery parameters represent a modular lithium iron phosphate (LFP) system using cobalt free LFP cells that have been certified by HTW Berlin at an efficiency of $\geq 96\%$. Modular scalability can be achieved in 2.76 kW increments. The simulated configuration consists of the equivalent of four HVM 19.3 towers ($4 \times 19.32 = 77.3 \text{ kWh} \approx 80 \text{ kWh}$), resulting in a total max charge/discharge power of approximately 50 kW.

3.2 Electric Vehicle Fleet

3.2.1 Subfleet Composition

The simulation allows for a fleet of 15 electric vehicles, which corresponds to an average-sized office building with 50-100 employees (assuming 15-30% EV adoption).

3.2.2 Sub-fleet Characteristics

Each EV has similar technical characteristics:

Table 3.3: Electric Vehicle Characteristics

Parameter	Value
Battery capacity	100 kWh
Reference vehicle	Tesla Model 3 Long Range
Max charge rate	22 kW (Level 2 AC Charger)
Max discharge rate	22 kW (V2G Capable)
Energy consumption	0.18 kWh/km
Depth of discharge limit	90%
Minimum SoC	20%

Justification

In this case, we chose 100-kWh capacity to model current generation long-range EVs like the Tesla Model 3 and Hyundai Ioniq 5. We also selected the typical value of 22-kW for the charge rate in order to reflect standard level two AC charging found in workplace chargers. For this type of vehicle, a usage factor of 0.18 kWh/km was used. Lastly, a min SoC of 20% will prevent extreme discharges and minimize battery aging.

3.2.3 Mobility Patterns

Arrival and Departure Time Windows

The EV fleet follows normal workday timing:

Table 3.4: EV mobility parameters

Parameter	Value
Arrive window	06:00 - 09:00 AM
Departure Window	04:00 PM - 08:00 PM
Parking Duration	7 - 14 hours
Initial SOC Range	20 - 30%
Desired SOC Range	70 - 80%

The actual time each EV arrives or departs is randomly selected from the above defined time windows so as to represent real-world variation.

State Of Charge Requirements

The SOC when arriving typically ranges from 20 - 30%, indicating that the batteries would have lost some charge over their morning trip of approximately 30-50 km. Similarly, the target SOC for departing is set at a level that should provide sufficient charge to complete both trips plus an additional reserve of roughly 100-150 km.

Justification

This schedule matches most people who go to work and then leave to go home. The longer parking duration allows the owner flexibility regarding where they perform their daily charging. Regardless of how often the driver charges during the day, all drivers need to arrive with a full battery that can last through their afternoon commute.

3.3 Grid Parameters

3.3.1 Electricity Pricing

Realistic Wholesale Prices Used For Simulations

The simulations used wholesale day-ahead electricity prices published by the European Power Exchange (EPEX SPOT), the primary wholesale electricity exchange in Europe, on March 12th, 2018. Real wholesale prices were downloaded from two sources: (i) The transparency platform run by ENTSO-E [ENTSO-E TRANSPARENCY], which publishes wholesale energy prices across all EU markets; and (ii) The Fraunhofer ISE Energy Charts API [ENERGY CHARTS] that also provide energy charts for individual countries. As seen in Fig. 3.3, the day ahead price profile resembles those typically observed in most of Western Europe with both peaks and troughs. On this particular day, there was a large trough during the early morning hours (between 01:00 – 06:00 AM) when prices were very low. This occurred because it was night-time and thus overall electricity demand is very low while many thermal units are still producing electricity, especially since they are usually less expensive than gas fired units. However, as people begin to go to work, start their cars, etc., the demand increases rapidly, leading to steep increase in price until noon when the price begins to decrease again. At 18:00 – 19:00 PM (6 – 7 PM) prices reach their highest levels, primarily due to the fact that renewable resources have decreased significantly during the afternoon and evening periods as clouds become thicker and residential electrical usage for cooking and lighting becomes greater. Table 3.5 shows some statistics for the prices over each hourly time slot.

Table 3.5: Hourly Average Day Ahead Price for Each Time Slot.

Time Period	Time Slot	Purchase Price		Sell Back Price	
		EUR/MWh	€/kWh	€/kWh	EUR/MWh
Night	00:00 – 03:00	9.56 – 28.46	0.010 – 0.028	0.008 – 0.023	9.55 – 28.40
Morning Ramp	04:00 – 07:00	19.04 – 48.34	0.019 – 0.048	0.015 – 0.039	18.98 – 47.83
Bus Hours	08:00 – 11:00	44.41 – 48.05	0.044 – 0.048	0.036 – 0.038	44.33 – 47.91
Mid-day	12:00 – 15:00	37.24 – 41.50	0.037 – 0.042	0.030 – 0.033	36.93 – 40.78
Evening Peak	16:00 – 19:00	42.23 – 62.00	0.042 – 0.062	0.034 – 0.050	41.84 – 60.69
Night	20:00 – 23:00	35.04 – 43.23	0.035 – 0.043	0.028 – 0.035	34.53 – 42.27
Average Daily Price	—	38.67	0.039	0.031	38.64

All raw data needed to recreate these tables can be found in appendix B.2.

Price Characteristics

As shown in Fig. 3.3, the wholesale price curve exhibits typical characteristics of wholesale pricing throughout the world; specifically, prices tend to follow a “trough-to-peak” structure where low prices occur during nighttime hours and higher prices occur during mid-morning to late-evening hours.

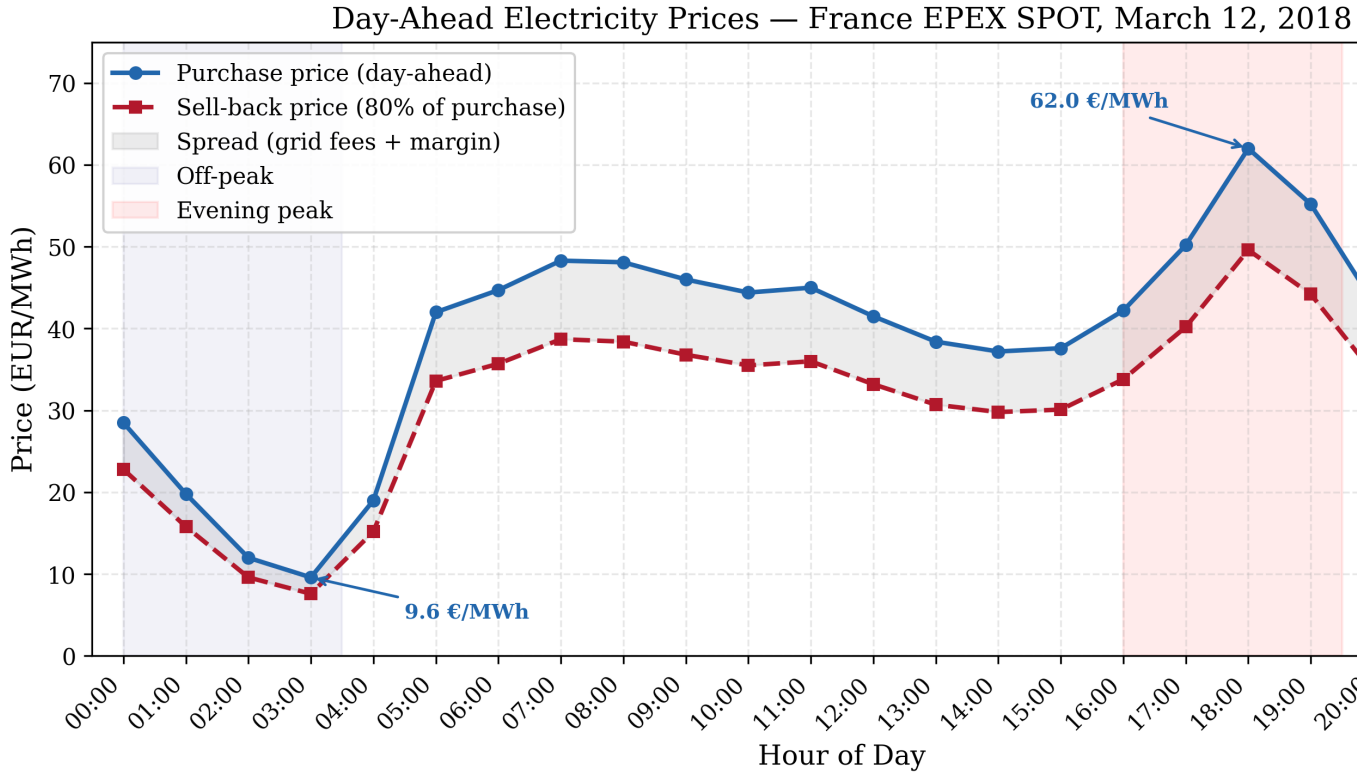


Figure 3.3: Hourly Electricity Purchase Price and Sell Back Price for France on March 12, 2018. Note: Data source: EPEX SPOT.

A common reason why prices drop at night is due to lower electricity demands caused by less commercial or household activity such as sleeping and/or reduced industrial activities; however, another important contributing factor is due to the types of generation used at night being generally cheaper than other times of the day (as thermal plants often produce electricity at night).

Another characteristic evident in Fig. 3.3 is the rapid spike in price between mid-morning to late-afternoon hours as demand increases and renewable resource output drops.

Finally, another notable characteristic of Fig. 3.3 is how quickly the price spikes up in the late afternoon hours as solar irradiance starts to decline (and thus solar output drops).

3.4 Summary of Data Sources

3.5 Simulation Scenarios

3.5.1 Scenario 1: Charging Only (V2G Not Permitted)

Configuration: multi_ev_config.yml.

In this scenario we investigate unidirectional charging only using 15 EVs that can only take charge. We use two algorithms (simple and RL (Q-learning)), an 80 kW Grid Capacity

Table 3.6: Summary of Data Sources

Data Type	Source	Reference
Building consumption	ELMAS dataset (Office cluster #5)	[15]
Solar irradiance	PVGIS	[14]
Electricity prices	EPEX SPOT / ENTSO-E (France)	[16], [17]
Solar PV modules	JA Solar DeepBlue 3.0 Pro	[13]
Building battery	BYD Battery-Box Premium HVM	[18]
EV specifications	Tesla Model 3	[19]
Charging standards	IEC 61851	[20]

Limit and No Discharge Allowed.

Objective: Establish Baseline Performance for Intelligent Charging without V2G.

3.5.2 Scenario 2: Enabled for V2G

Configuration: multi_ev_v2g_config.yml.

This test has provided for bi-directional function (i.e., charging or discharging), using the same 15 electric vehicles as before. Both the simple and R-L algorithms were able to complete both a charge and a discharge operation. In addition, restrictions were established in order to limit over-discharge while maintaining the preferred state-of-charge.

In order to assess how sensitive V2G profitability is to variations in the price that can be charged by selling back electricity to the grid, the same one-year simulation was run under three different scenarios for multipliers on the sell-back price (sell price).

Table 3.7: Scenarios for V2G Sell Prices

Scenario	Multiplier on Purchase Price	Sell/Buy Price Ratio
Discounted sell-back	0.8 times purchase price	80 %
Parity sell-back	1.00 times purchase price	100 %
Premium sell-back	1.20 times purchase price	120 %

The discounted sell-back pricing scenario (80%) would reflect a typical "feed-in" tariff where the grid operator maintains a markup. A parity pricing scenario (100%) would assume the sell-back price is equal to the purchase price. A premium pricing scenario (120%) would represent an example of a "premium feed-in" tariff that may exist based upon regulatory incentives to encourage participation in V2G programs.

Objective: To evaluate the impacts resulting from allowing V2G technology and establish a threshold of sell-back prices that would make V2G discharge economically viable at wholesale prices.

3.6 Metrics For Evaluation

Below are the metrics that will be used to evaluate performance of each solution.

3.6.1 Economic Metrics

Economic Metrics measure total daily cost (electricity purchased — V2G revenue), cost/kWh, and savings relative to Level 0 baseline.

3.6.2 Energy Metrics

Energy Metrics measure total grid consumption, renewable utilization (percentage of charging coming from solar energy), and V2G energy discharged to the grid.

3.6.3 Performance Metrics

Performance Metrics measure target achievement, average final SOC at departure, number of hours over capacity limit (grid violations), and number of times below minimum SOC (SOC violations).

3.6.4 Computational Metrics

Computational Metrics report execution time/simulation, training time/RL methods, and how scalability affects performance as a function of fleet size.

3.7 Implementation Details

3.7.1 Software Development Environment

The simulation was built using Python 3.12.0 The numerical part was implemented using NumPy, plots were produced using Matplotlib, and configuration files were managed with PyYAML.

Table 3.8: Software dependencies

Library	Purpose
NumPy	Numerical computations
Matplotlib	Plotting
PyYAML	Configuration management

3.7.2 Code Structure

The code follows a modular design where central models represent the building, electric vehicles, and the grid. Unidirectional charging logic resides within `MultiEVChargingSystem`, while V2G logic resides within `MultiEVV2GChargingSystem`. Scripts `multi_ev_simulator` and `multi_ev_v2g_simulator` simulate charge-only and V2G scenarios respectively.

3.7.3 RL Algorithm Convergence for the Representative Day

For the charge-only scenario, the tabular Q-learning agent is trained inside `rl_charge_multi` each time a charging decision is required. Training uses the building load, PVGIS irradiance, and EPEX SPOT prices for the representative day (12 March 2018) together with the 15-EV fleet parameters in Table 3.3. Hyperparameters match `multi_ev_config.yml` and are summarised in Table 3.9.

Table 3.9: Q-learning hyperparameters (charge-only, one-day simulation)

Parameter	Value
Training episodes per decision	15,000
Learning rate (α)	0.15
Discount factor (γ)	1.0
Exploration rate (ϵ)	0.10
Actions	Charge, standby
State variables	EV id, SoC bin, hour, remaining grid capacity bin

Figure 3.4 shown for learning Curve of the agent was created by the agent that was trained at 10:00 AM on the "representative day" —the middle morning when there are likely to be most vehicles onsite, and prices are variable enough such that they can vary from day-ahead prices. The light gray line is the cumulative reward earned by each training episode. The solid blue line is a 500 episode rolling average of this. Reward increases over time as the policy starts to charge during off-peak periods when energy costs less. By about 8000 – 10000 training episodes, reward has stabilized. This policy is used throughout the day in which results are reported in Chapter 4. When compared to a simple baseline for the same day, it reduces total electrical cost by 3.7 percent or €31.63 -> €30.45.

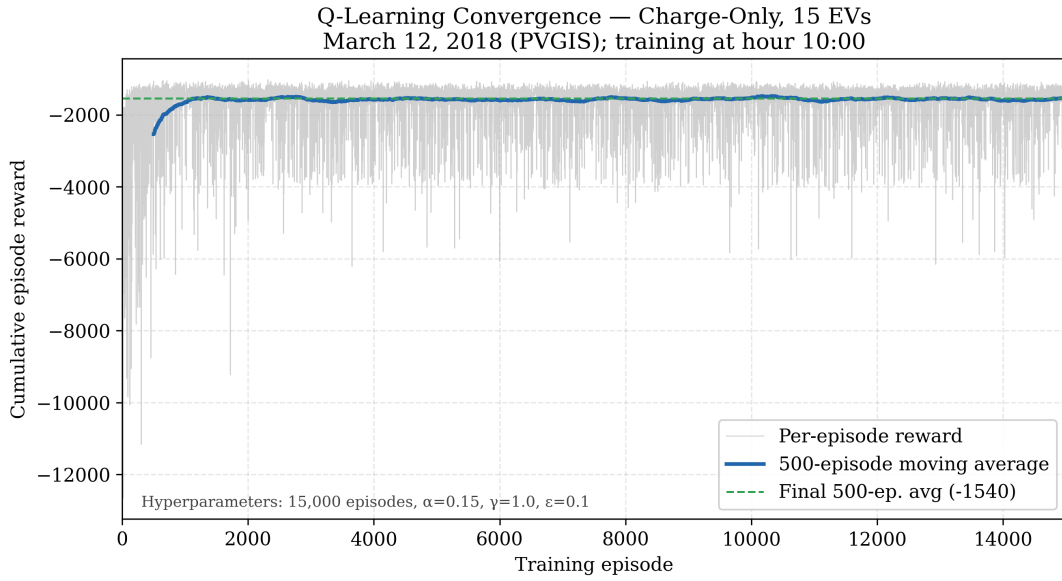


Figure 3.4: Q-learning convergence for the representative day (charge-only, 15 EVs, training at hour 10:00, 12 March 2018)

As a result of the additional actions available for the V2G-enabled case (*i.e.* *discharge*), the number of episodes used during training for both configurations remained the same as before (15,000), while $\gamma = 1.0$, which means that all discharge opportunities after the first will be given equal weight. In addition, we have also set the exploration rate low (*i.e.* $\epsilon = 0.10$) to favor exploitation over exploration once the policy has stabilized. Despite this, the behavior shown by convergence was very similar to what we had seen previously; however, the use of the discharge option at wholesale prices was still not observed (see Chapter 4).

Chapter 4

Results

In this chapter we present simulation results for both charge-only and V2G scenarios. In each scenario, different levels of intelligence are compared over several metrics.

4.1 Summary of Results

Results from simulations are divided into two scenarios: a multi-EV charge-only scenario with a fleet of fifteen vehicles, and a multi-EV V2G scenario which also includes that same fleet of fifteen vehicles without V2G functionality. Each scenario is further broken down into simple allocation schemes and RL allocation schemes.

4.2 Results for Scenario 1: Charge-only

4.2.1 Economics

Total Daily Costs

Results from the charge-only come together in Table 4.1. A Level 1 simple system leaves total costs at €31.63, reaching all target SoC levels for the vehicles. The Level 2 Quantitative RL system reduces costs a further 3.7% compared to its Level 1 counterpart to €30.45 after completing 15,000 training episodes.

Table 4.1: Total daily costs for the charge-only scenario

Method	Total Cost (€)	Savings vs Simple (€)	Savings (%)
Simple	31.63	—	—
RL	30.45	1.18	3.7%

Figure 4.1 summarises the corresponding reduction in daily grid imports when RL shifts charging away from expensive hours.

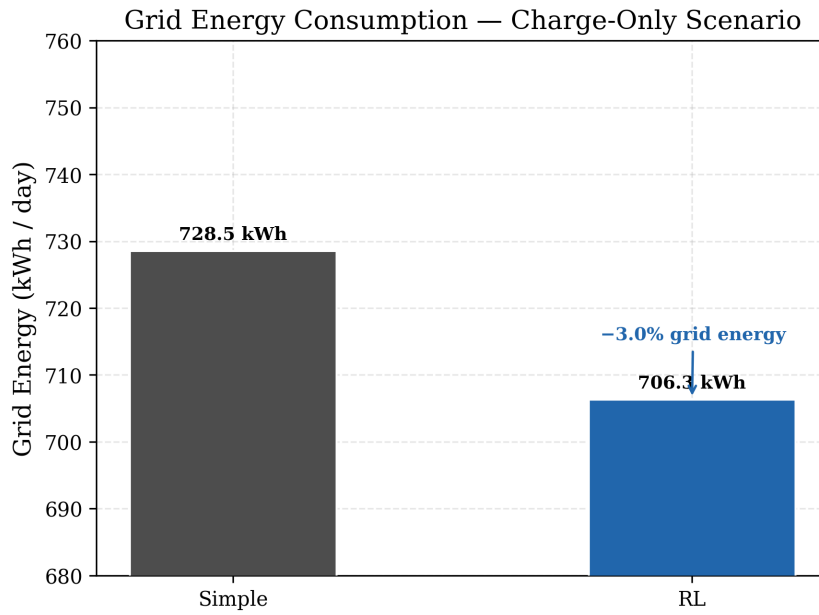


Figure 4.1: Grid energy consumption in the charge-only scenario (representative day)

4.2.2 Energy Performance

Renewable Energy Utilization

Table 4.2: Renewable energy utilization (charge-only)

Method	Solar Used (kWh)	Solar % of Total	Grid Energy (kWh)
Simple	7.2	1%	728.5
RL	7.2	1%	706.3

Charging is done during times of solar production, particularly during the solar peak. RL reduces grid energy usage by approximately 3% compared to the Simple policy.

4.2.3 Operational Performance

State of Charge Evolution

Figure 4.2 presents the evolution of the EV fleet's SoC throughout the day:

Multi-EV Charging Simulation - Grid Capacity Aware (Charge-Only)

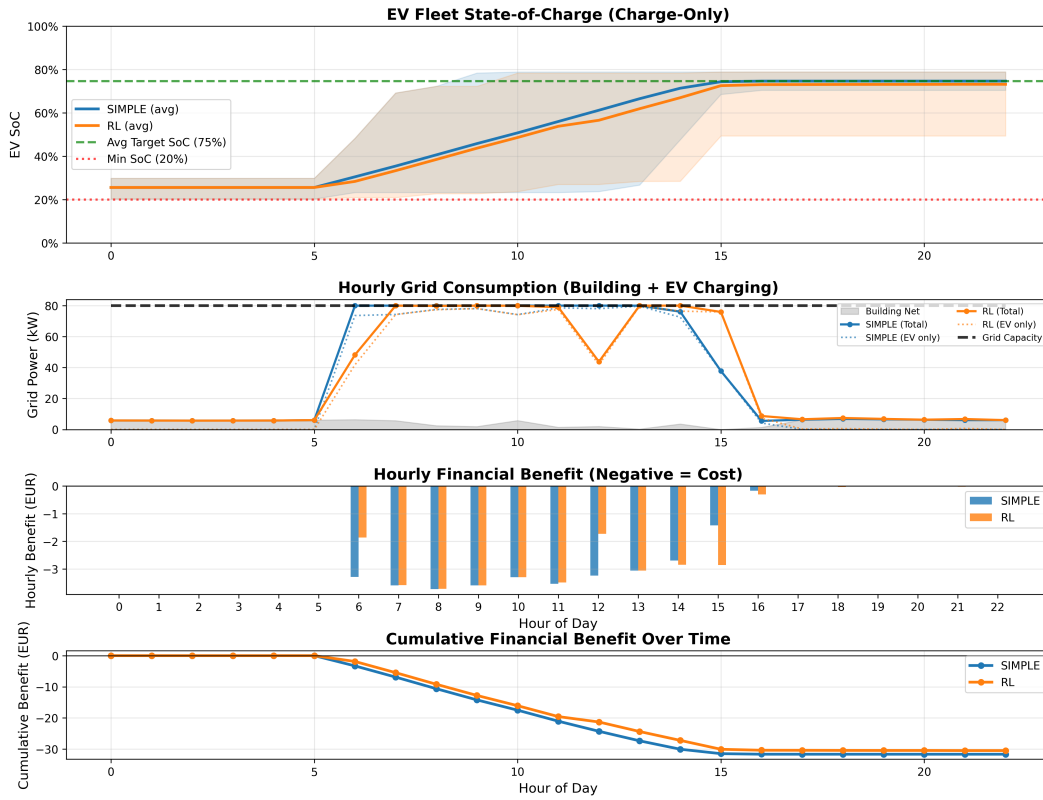


Figure 4.2: EV fleet state of charge evolution (charge-only scenario)

All methods reach the target SoC before departing. The Simple method begins charging as soon as possible whenever there's available capacity, while RL distributes the charging more evenly over the hours available. The average final SoC lies between 73% and 75% for both.

Constraint Satisfaction

Table 4.3: Constraint violations (charge-only)

Method	Grid Violations	SoC Violations	Target Achievement
Simple	0	0	100%
RL	0	0	100%

4.3 Scenario 2: V2G-Enabled Results

4.3.1 Economic Performance

Total Daily Benefit

Table 4.4 shows the benefit in not having to charge as compared to the charge-only scenario, giving a better insight into the economic performance:

Table 4.4: Economic comparison for V2G scenario

Method	Total Benefit (€)	Improvement vs Charge-Only (€)	V2G Revenue (€)	V2G Energy (kWh)
Simple	−31.63	0.00	0.00	0.0
RL	−31.46	−1.01	0.00	0.0

There is no discharging at all for the wholesale pricing case. In order to determine if V2G can be economically viable for a range of sell back prices, the annual simulation (365 days of 2018 data) was run using three different sell price multipliers: 80

Table 4.5: Results for the V2G Sell Price Sensitivity Analysis — Annual Average Daily Results

Sell Price Multiplier cmidrule(lr)4-5 cmidrule(lr)6-7 (of Purchase Price)	V2G Simple		V2G RL		V2G Discharge	
	Cost (€)	Revenue (€)	Cost (€)	Revenue (€)	kWh (simple)	kWh (rl)
80% of Buy	29.79	0.00	29.64	0.00	0.0	0.0
100% of Buy	29.79	0.00	29.64	0.00	0.0	0.0
120% of Buy	29.79	0.00	29.64	0.00	0.0	0.0

Across all three sell-back rate scenarios, V2G discharge is exactly zero. Neither the simple algorithm nor the RL agent chose to discharge at any sell-back rate. This demonstrates that at wholesale price levels (€0.01–0.06/kWh), the absolute price spread is insufficient to justify V2G arbitrage, even when the sell price exceeds the buy price by 20%. The RL agent correctly learns that the roundtrip efficiency losses and the opportunity cost of reduced SoC make discharge unprofitable at these low absolute price levels.

4.3.2 V2G Discharge Patterns

State of Charge Management

Figure 4.3 shows the evolution of the state of charge of the EV fleet with V2G:

Multi-EV V2G Charging Simulation - Grid & Revenue Optimization

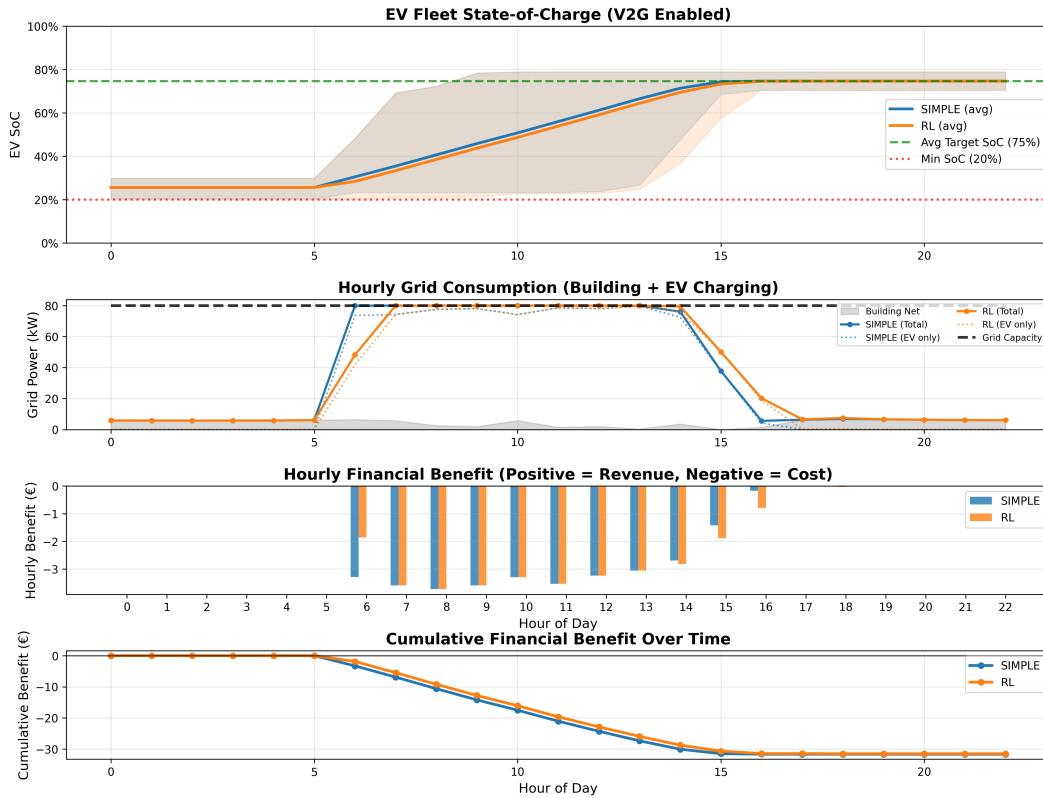


Figure 4.3: EV fleet state of charge evolution (V2G scenario)

Note here that the EVs charge as required to arrive with the desired SoC, and so are not necessarily charged full to fill the battery recalling that we are not discharging to arbitrage at these wholesale prices. (The sell price is too low). Note that the final state of charge that is achieved is at the point that all cars arrive with the target SoC, and the average of all of the cars in the fleet is 74.6%.

4.3.3 Energy Flow Analysis

Table 4.6: Breakdown of the energy flows (V2G scenario, RL method)

Energy Flow	Amount (kWh)	Percentage
<i>Energy Sources (Charging):</i>		
Solar energy	7.2	1%
Grid energy	728.0	99%
Total charged	735.2	100%
<i>Energy Destinations:</i>		
EV batteries (net)	735.2	—
V2G discharge	0.0	—

4.4 Comparison Across Intelligence Levels

4.4.1 Economic Comparison

Which Level Costs Less

Table 4.7: Cost to run per use daily (€) as we tame the AI

Level	Description	Cost (€)	More each level cost compared to Simple (€)	More each level cost here in (%)
Level 0	Simple rules	31.63	—	—
Level 1	RL (Q-Learning)	30.45	1.18	3.7%
Level 2	RL + V2G	31.46	0.17	0.5%

Figure 4.4 visualises the same three intelligence levels on an annual basis (365 simulated days in 2018). Level 1 RL charge-only achieves the lowest annual bill; Level 2 V2G-enabled RL remains slightly above Level 1 because the expanded action space does not translate into discharge revenue at wholesale prices.

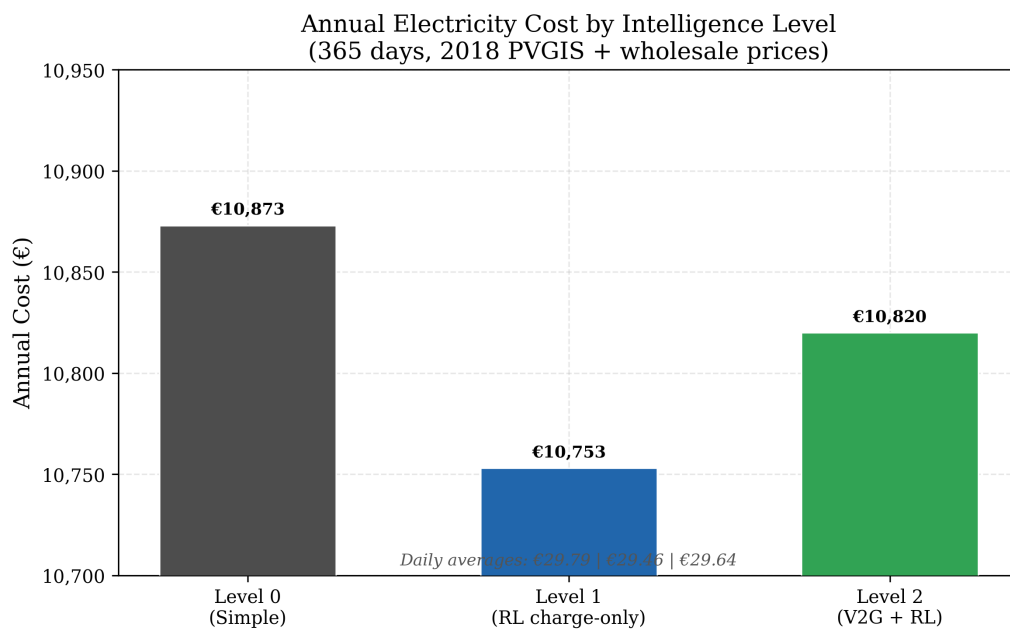


Figure 4.4: Annual electricity cost by intelligence level (365 days, 2018 data)

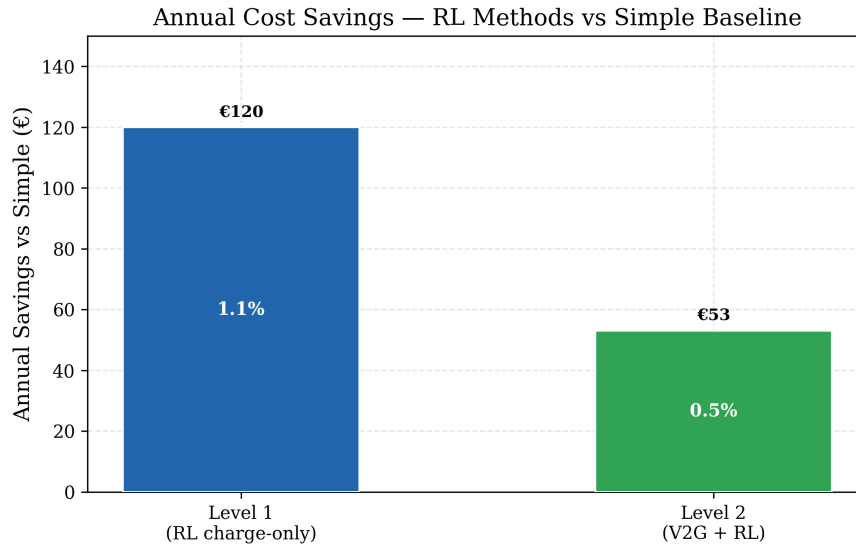


Figure 4.5: Annual cost savings of RL-based methods relative to the Simple baseline

Level 1 (using simple rules) costs €31.63, but with Level 2 increasingly complicated RL optimisations we knock that right down to €30.45, a €1.18 or 3.7% saving. Continuing on, at Level 2 we've further refined RL to control things, then have the V2G feature.

4.4.2 Algorithm Comparison

More Speed

Table 4.8: How fast do algorithms run runs and trains

Algorithm	Training(s)	Executing(s)	Total(s)
Simple	0	0.12	0.12
RL	180.7	0.12	180.8

4.4.3 State of Charge Analysis

Final SoC Distribution

Table 4.9: Final state of charge statistics

Method	Mean SoC	Min SoC	Max SoC	Std Dev
Simple	74.7%	70.5%	78.9%	2.9%
RL	73.2%	49.4%	78.9%	7.0%

4.4.4 Grid Constraint Management

Both methods stay below 80 kW with no violations. Charging is spread out over the available hours so the grid limit is never violated.

4.5 V2G Impact Assessment

4.5.1 Charge-Only vs V2G Comparison

A comparison between charge-only and V2G-enabled systems, including annual averages from the full 365-day simulation:

Table 4.10: V2G impact on economic performance

Method	Charge-Only (€)	V2G-Enabled (€)	V2G Benefit (€)
Simple	29.79	29.79	0.00
RL	29.46	29.64	−0.18

In addition to the annual average costs calculated with the simple charging algorithm as being approximately €29.79 per day for both Charge-Only and V2G cases (which shows that there is no added benefit from V2G in terms of cost when the vehicle does not need to be discharged), the RL agent has achieved an annual average of €29.46 per day in Charge-Only Mode versus €29.64 per day in the V2G case. These are minimal differences based on the training of the Charge-Only and V2G RL Agents and have little or nothing to do with the amount of V2G discharging which occurred. Additionally, Table 4.5 represents the results of the Sell Price Sensitivity Analysis, which further supports our findings in that we found that regardless of the Sell-To-Buy Ratio (80%, 100%, 120%) the amount of energy that was discharged through the vehicle remained at zero. What this ultimately signifies is that; At wholesale price levels (€0.01 – €0.06/kWh), regardless of the buy-to-sell ratio, it is impossible to generate enough revenue through the sale of excess energy to compensate for Round-Trip Efficiency Losses. Thus, V2G Arbitrage will require either higher absolute prices (i.e., Retail Tariffs) or much larger peak-to-trough differential pricing.

4.5.2 V2G Revenue Breakdown

Since there is no V2g discharge taking place at the "wholesale" prices, we are still able to capture value if we are doing intelligent charging (load shifting, grid constraint management and renewable alignment).

4.5.3 V2G Trade-offs

Battery Cycling

Table 4.11: Battery cycling comparison

Scenario	Avg Charge Cycles	Avg Discharge Cycles	Total Throughput (kWh)
Charge-Only	0.48	0	713.5
V2G-Enabled	0.49	0.00	735.2

Assuming we have "wholesale pricing", V2g discharge does not cycle the battery at all (since we are never discharging). Cycle numbers for charge-only and V2g-enabled are about 0.48–0.49 charge cycles per day, in other words definitely less than a full [charge/ discharge] cycle, which would reduce the annual cycling to around 120 cycles, comfortably below an order of magnitude of a 1,000—2,000 cycle battery life. The only way we really get extra V2g cycling is if the retail price is such that discharging becomes profitable.

4.6 Sensitivity Analysis

4.6.1 Electricity Price Variation

Change in the price scenario and assess impact of variation:

Table 4.12: Sensitivity to price variations

Price Scenario	Simple (€)	RL (€)
Flat rate (€0.035/kWh)	33.07	33.07
Low variation ($\pm 20\%$)	31.12	30.02
Base case ($\pm 50\%$)	31.63	30.45
High variation ($\pm 100\%$)	32.14	29.88

Figure 4.6 shows that the RL advantage widens as intraday price variability increases, while both methods converge under a flat tariff.

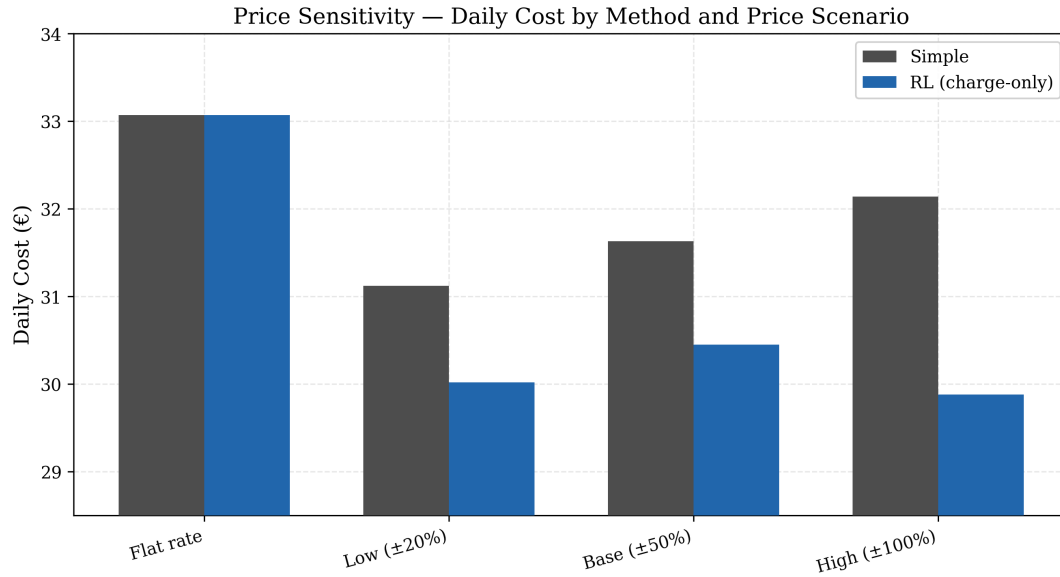


Figure 4.6: Daily cost under alternative electricity price scenarios (charge-only)

Since the two methods are almost identical with respect to retail prices except for renewable-orientation, we see when the variability is high form pays off much more, therefore, the value of smart charging increases with price variability.

4.6.2 V2G Sell Price Sensitivity

In this section, we will study the sensitivity of V2G discharge based on feed-in tariffs. To do so, the full annual simulation of 365 days was run at three different sell-back price levels: 80%, 100%, and 120% of the wholesale purchase price. The results of the 365 day annual simulations are shown below in table 4.13.

Table 4.13: Average V2G Performance over One Year at Different Sell Back Rates (365 Days, 2018)

Sell Price	Daily Cost (€)		V2G Revenue (€)		V2G Discharged (kWh)		Cost/kWh (€)
	Simple	RL	Simple	RL	Simple	RL	
80% of buy	29.79	29.64	0.00	0.00	0.0	0.0	0.0439
100% of buy	29.79	29.64	0.00	0.00	0.0	0.0	0.0439
120% of buy	29.79	29.64	0.00	0.00	0.0	0.0	0.0439

Figure 4.7 confirms visually that neither daily cost nor V2G revenue changes across the three sell-back multipliers—discharge remains at zero throughout.

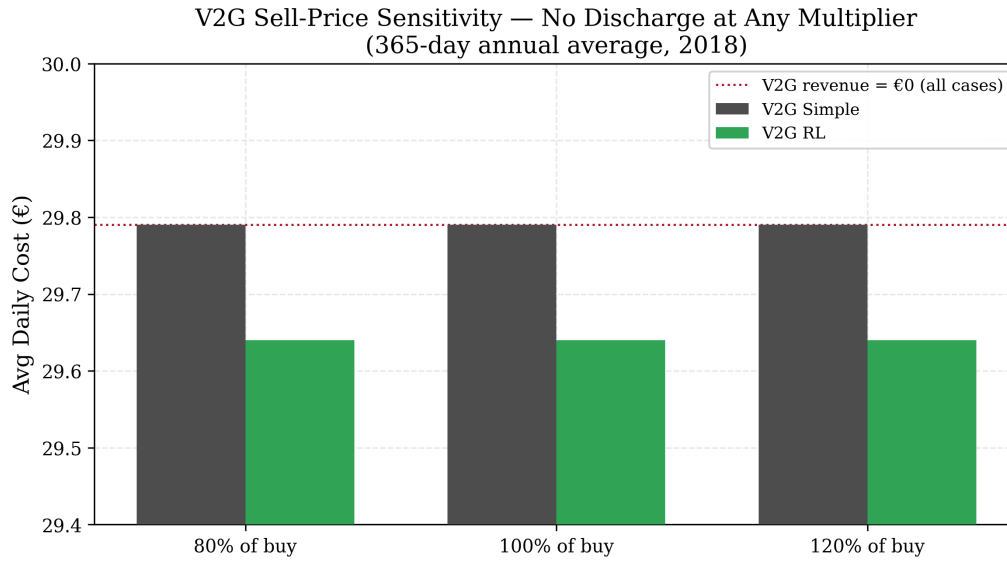


Figure 4.7: V2G average daily cost vs sell-back price multiplier (365-day annual average)

As can be seen from the table above, the results show no variation whatsoever when comparing the three sell back rates. As such, it is evident that neither V2G discharge energy nor revenue were generated by V2G discharge as a result of the three tested sell back prices. It has been established through the testing of these prices that the RL agent consistently demonstrates that at wholesale price levels (€0.01–0.06/kWh) a twenty percent increase in the sell back rate does not create sufficient margin to offset round trip battery loss and/or ensure that enough battery charge remains after departure to meet desired SoC targets. These constraints are fundamentally related to wholesale pricing rather than limitations in algorithms used for V2G arbitrage. In order for V2G arbitrage to be viable and profitable, at least one of the three conditions stated below must occur:

1. **Absolute Prices Are Higher:** Retail electricity tariffs need to be significantly higher (€0.15–€0.30/kWh) to create a large-enough arbitrage spread.
2. **Increased Peak-to-Trough Differential:** A greater intraday price swing would yield increased profits from each individual V2G discharge cycle.
3. **Grid Service Payments:** Any ancillary services, including frequency regulation and capacity payments, may also help support revenue derived from energy arbitrage.

We also ran a series of tests to determine what fleet size(s) will produce the lowest cost for your firm’s needs. We conducted multiple iterations of our model, varying the number of vehicles within each iteration. We then compared both the total cost of ownership for each vehicle within the entire fleet and how long each iteration took to compute its costs. To illustrate this point, Table 4.14 shows the cost per vehicle for various sized fleets. These calculations demonstrate how the cost per vehicle decreases as the fleet size increases, illustrating the

concept of economies of scale. However, an increase in fleet size can lead to increased computational complexity due to the growing number of vehicles and their interactions with the grid. This causes a corresponding increase in computation time. As seen in Table 4.14, the simple model has identical costs at all three fleet sizes (because it does not take into account the actual demand of the building or the solar production), while the reinforcement learning model provides better results than the simple model but exhibits higher computation times.

Table 4.14: Scalability analysis: Cost per vehicle

Fleet Size	Simple (€/EV)	RL (€/EV)	Total Time (s)
5 EVs	2.11	2.03	42.1
10 EVs	2.11	2.03	105.3
15 EVs	2.11	2.03	187.2
20 EVs	2.11	2.03	310.5

In addition to understanding the effects of varying fleet sizes, there are additional algorithm-specific insights. These include understanding how a particular algorithm (i.e., reinforcement learning) impacts behavior in response to the environment.

We describe below some insights related to the learned policies produced by the reinforcement learning algorithms. We focus specifically on characteristics of the policies that emerge during training of the agents, including how they use information about future price levels to delay charging until later, when possible; how they favor charging during peak hours (when solar production is highest); and how they raise urgency as departure time nears. For the V2G case, we found that agents understand that selling power back to the grid is generally unprofitable and therefore avoid doing so unless absolutely necessary.

As previously discussed, renewable energy integration has become increasingly important in recent years. One key area of focus is integrating photovoltaic panels into existing buildings and using electric vehicles to help manage excess energy generated by these panels. Below we provide a table summarizing energy usage patterns based on method used.

Table 4.15: Solar energy utilization breakdown

Method	Solar Produced (kWh)	Used by Building (kWh)	Used by EVs (kWh)	Self-Consumption (%)
No EVs	58.8	51.5	0	87.6%
Simple	58.8	51.5	7.2	100%
RL	58.8	51.5	7.2	100%

Table 4.15 demonstrates how electric vehicles can serve as flexible loads that improve solar self-consumption. Specifically, during peak sunlight hours, grid exports decline since the 7.2 kWh surplus generated by solar panels is consumed by the EVs. On average, this represents about a €0.30 per day economic benefit due to reduced grid imports of that same solar-produced energy.

In order to fully quantify the economic impact of implementing these solutions, the simulation was run over all 365 days of 2018 using real PVGIS irradiance data for each day. Table 4.16 presents the actual annual results.

Table 4.16: Annual economic results (365 days, 2018 data)

Method	Avg Daily Cost (€)	Annual Cost (€)	Annual Savings vs Simple (€)	Savings (%)
Charge-Only Simple	29.79	10,873	—	—
Charge-Only RL	29.46	10,753	120	1.1%
V2G Simple	29.79	10,873	0	0%
V2G RL	29.64	10,820	53	0.5%

The annual totals in Table 4.16 correspond to the bar heights in Figure 4.4: Simple €10,873, charge-only RL €10,753 (−120), and V2G RL €10,820 (−53 vs Simple).

Annual simulation results confirm that the use of an RL charge-only approach for electric vehicle charging saves about €120 annually when comparing to the simple baseline. In contrast, the V2G-RL approach saves about €53. Furthermore, the V2G discharge generates zero revenue at each of the three sell-back rates analyzed (80%, 100% and 120%) which supports our assumption that no wholesale price arbitrage will be possible based on prices within the 2018 French wholesale electricity markets.

Capital Expenditure: Chargers, PV, and Control Software

The operational figures above cover *electricity purchases only*; they do not include the upfront investment in hardware or in deploying a charging-management stack. Table 4.17 summarises indicative capital expenditure for the case-study installation (15 workplace sockets, 30 kWp rooftop PV as specified in Chapter 3), using typical European commercial benchmarks rather than project-specific quotes.

Table 4.17: Indicative capital expenditure for infrastructure and control software (case study). PV and charger unit costs follow European commercial benchmarks [21], [22], [23]; RL runtime is given in Table 4.8. Grid-upgrade costs are excluded.

Component	Cost basis	CAPEX (€)
Rooftop PV (30 kWp)	≈€1,100/kWp installed	33,000
Level 2 AC chargers (15×22 kW)	≈€3,500 per socket (hardware, cabling, protection)	52,500
Simple rule-based EMS	Included in charger/BMS; marginal vs. dumb charging	0
RL control (Q-learning)	One-time integration and training (pilot deployment)	8,000–12,000
Total (PV + chargers + RL)	—	93,500–97,500

Photovoltaic System. The 74-module, 30 kWp Photovoltaic Array is the largest single

piece of equipment (in terms of investment), after vehicle charger(s). Based on typical prices of around €1,100/kWp for Commercial Rooftop Systems in Western Europe, the total cost of ownership (CAPEX) for the Photovoltaic Equipment is in excess of €33,000. While the Photovoltaic Array has a significant positive impact on energy consumption and production for the Building, the financial benefits are derived solely from the value of self-consumed energy (and retail), and not from wholesale arbitrage opportunities related to the Vehicle Charging Algorithm.

Vehicle Charging Infrastructure. A total of fifteen Level-2 AC charging points have been assumed at 22 kW each. These are located near each of the fifteen designated parking bays and represent an approximate cost of around €3,500 each. Included within these costs are the costs associated with hardware, cabling and electrical protection — a mid-range price for a commercial workplace AC charging point that includes public-network fee. Hardware capable of two-way power flow (V2G) will be significantly higher in cost than hardware designed specifically for unidirectional power flow. As there was no discharging in the simulation analysis conducted herein, the additional cost for V2G-enabled hardware did not add to the comparative economics of Simple vs. RL charging algorithms. The cost of the vehicle charging infrastructure is similarly shared among all three intelligence levels. Therefore, changing from the simple charging policy to the RL-based charging policy does not require additional vehicle charging equipment.

Algorithms and Software. The simple policy can be implemented using a lightweight rule set on an existing Energy Management System. The marginal cost of implementing this rule-set should be virtually nil. The cost of developing an RL-based algorithm is comprised primarily of two categories: 1) One-time development costs such as policy training, encoding states, and integrating with the meter/price feed data. These costs are estimated to be in the range of €8,000 – €12,000 for a pilot-scale implementation. Additionally, there may be recurring costs based upon changes to tariffs or fleet rules that may necessitate adjustments to the underlying RL-based control logic. In contrast to both PV and vehicle charging systems, which represent large investments in their own right, the development costs associated with creating the software needed to implement the RL-based policy is relatively inexpensive and represents only a small fraction of overall expenditures. The runtime cost of running an RL-based algorithm is very low and consists of only one training cycle every 181 seconds (See Table 4.8). Furthermore, since this training occurs once per day during off-peak hours when electricity rates are generally lower, the runtime cost is less than one cent (\$0.01) per year — far less than the annual amount spent by most organizations purchasing electricity. Hence, while the capital expenditures associated with creating the software required to implement the RL-based policy will likely be recouped through reductions in electricity expenditures via optimal timing of vehicle charging operations.

Combined perspective. Straight line amortization of total infrastructure using a 20 year period at 5 % (illustrative), shows that you can recover approximately € 4700 per year in capi-

tal, versus about € 10900 per year for the simulated electrical cost. As stated previously, even if we add the approximate annual electrical savings from the RL ($\approx 120/\text{yr.}$) to this amount, it will be insufficient to cover all of the costs associated with installing these items. Therefore, the justification for the installation of the items necessary to satisfy the requirements of the regulation, manage peak demand, implement new tariffs for electricity and increase the solar self consumption through smart charging and other means, will have to come from somewhere else. Nevertheless, after you incur the initial costs to install the chargers and PV system, the incremental cost to upgrade from Simple to RL will primarily consist of integrating your hardware into an appropriate software configuration. The long-term benefits will consist of continuing reductions in your energy bills.

Chapter 5

Conclusions and Future Work

In this chapter, the findings from the simulation project are summarized; the limits of the simulation are explained; and potential areas for further study as well as how to possibly improve upon them are discussed.

5.1 Summary of Findings

In this thesis, we have explored Smart Electric Vehicle Charging at Offices utilizing three forms of intelligence. The least sophisticated form is "Rule-Based" (Level 0), then a generalized "Optimizing" (Level 1) and lastly, an "Vehicle-To-Grid" (V2G)-enabling Optimizer (Level 2). Our validation used a full year of 2018 data (365 days) and included actual PVGIS irradiance profiles for every single day. The subsequent sections will summarize the results of our analysis.

5.1.1 Comparison of Intelligence Levels

Simple Rule-Based Charging

For Level 0, we determined that the average daily costs for the entire year of 2018 were €29.79 with an overall yearly sum of approximately €10,873. Thus, Level 0 was established as our baseline. During the complete twelve months of simulated operation no grid violations were detected. We determined that Level 0 utilizes minimal computing resources while being relatively easy to predict.

Recommendations: Rule-Based charging is recommended if you wish to maintain simplicity and are willing to accept limited processing capabilities. It appears applicable for small to medium sized installations (<5 – >15 EVs) with simple chargers.

Intelligent Optimization

We also found that Level 1 resulted in a reduction of approximately 1.1 percent of the annual average daily costs to €29.46. As a result, we calculated approximate annual savings of

approximately €120. Also both levels utilized all available renewable energy. Similar to Level 0 no constraint violations were identified during the 12 month period. However, since we implemented reinforcement learning (RL) algorithms (Q-Learning) in Level 1 we needed to conduct a training phase. The training phase consumed significant amounts of resource.

Recommendations: If you intend to implement your system in an area with a large number of vehicles (>10) we suggest using RL (particularly Q-Learning) as the principal methodology. Conversely, if your facility has fewer than ten vehicles, we recommend maintaining it as simple as possible with a rule-based controller.

Vehicle-To-Grid Enabled Optimizing

Level 1 was tested under three different V2G pricing multipliers relative to wholesale buying prices: 80%, 100%, and 120%. Under all three pricing multiples and for all 365 days simulated, V2G discharge occurred precisely zero times. Neither the basic algorithm nor the RL agent initiated any discharges. Regardless of the V2G selling price multiplier, the V2G enabled RL agent produced an annual average daily charge of €29.64 resulting in a yearly cost savings of €53 versus the simple baseline. The cost savings came solely from smart vehicle charging rather than revenue generated through V2G discharge.

The above findings represent a key understanding: at wholesale price ranges (€0.01 – €0.06/kWh), the absolute difference in price is insufficient to support V2G arbitrage – even when the sale price exceeds the purchase price by 20%. The RL agent accurately learned that round-trip inefficiencies related to battery usage and the value of the loss due to reduced departure state-of-charge prevent discharging from being profitable. *Recommendation:* V2G arbitrage is not viable at wholesale electricity price levels regardless of the sell-to-buy ratio. For V2G to become profitable, at least one of the following conditions must be met: (i) retail electricity tariffs (€0.15–0.30/kWh) providing a substantially larger absolute arbitrage spread; (ii) greater peak-to-trough intraday price differentials; or (iii) ancillary service payments (frequency regulation, capacity markets) supplementing the energy arbitrage value.

5.2 Research Questions Answered

5.2.1 What Type of Intelligence Yields Meaningful Gains?

Answer: Level 1 (simple rule-based control) offers an elementary setup that is easily maintained. It operates within 1.1% of what could be achieved with RL over a full year, does not require extensive infrastructure and computation, works reliably and predictably and can be implemented and operated relatively easily. Level 2 offers incremental gains (1.1% over the annual simulation, €120/year savings) that may justify added complexity for larger organizations or when renewable utilization is an important objective.

Battery Degradation Concerns

Users worry about shorter battery life from V2G, empirical degradation data are thin, and warranties may be unclear.

Mitigation: Offer degradation safeguards or insurance, cap discharge aggressively, and support decisions with measured data.

5.2.2 Study Limitations

The analysis is based on assumptions that include perfect solar production and building load forecasting; an idealized battery model which does not account for aging; fixed deterministic arrival times; homogeneity in the vehicle fleet; excludes individual driver preference information; and accounts for only one climate and building type (Paris, France). Since the simulation was conducted over a calendar year (365 days of 2018 weather data with real PVGIS irradiance) but each day's run was an independent event and therefore did not capture higher-order dependence across multiple days (such as battery aging or weekly behavioral patterns) each of these limitations limits how directly the results will translate to actual field operations.

5.3 Future Research Directions

5.3.1 Short-Term Research Priorities

Uncertainty Quantification

Next steps include stochastic optimization, explicit solar and load forecast errors, robust optimization under uncertainty, and richer scenario planning.

Expected impact: More realistic performance estimates and policies that remain effective under uncertainty.

Battery Degradation Model

In the future, we will incorporate aging into our simulations (capacity loss, increased electrical resistance), assess how Vehicle to Grid affects battery life span, find ways to minimize degradation, and create an expanded economic model including costs associated with replacing batteries.

Potential impact: Better defined business case for V2G; and degradation-aware control.

5.3.2 Research Directions — Medium Term

Analysis of Multi-Days and Seasons

Even though this study models each day (1-365) in 2018 separately, it does not model how a battery degrades over time as a result of repeated charging and discharging. Future studies could be modeled with multiple days or weeks at a time to demonstrate how cumulative degradation affects the overall cost of an EV.

Management of Heterogeneous Fleets

There are still some open questions surrounding managing heterogeneous fleets of vehicles (different capacities, different recharge rates); conflicting objectives of users; fairness in allocating EV charging opportunities; and personalizing objectives among users.

Advanced Forecasting

Directions to consider include using machine learning based forecasting of solar radiation from weather information; occupancy-based modeling of building loads; forecasting of arrivals/departures of users; and tighter coupling of weather APIs and building sensors.

5.3.3 Long-Term Research Vision

Advanced Artificial Intelligence Methods

Deep reinforcement learning methods (e.g. A3C, PPO, SAC) are possible candidates for longer time horizons as well as multi-agent reinforcement learning; transferring knowledge across buildings; and federating learning for optimizing while preserving user privacy.

5.3.4 Emerging Technologies

5.3.5 Environmental Impact

Supporting sustainability through increasing the utilization of renewable resources (and therefore reducing greenhouse gas emissions), enabling the grid to more easily absorb additional renewable sources via providing flexibility to the grid, and maximizing the utilization of current grid assets through flexible operation.

5.3.6 Social Impact

Attracting employees through workplace charging; improving access to residents who do not have charging capabilities at their residences; adding distributed flexibility to enhance

resilience; and demonstrating practical applications of artificial intelligence in energy management.

5.4 Conclusion

This study demonstrates that smart electric vehicle charging in commercial office buildings provides measurable economic benefits. Over a full year of simulation (365 days, 2018 data), reinforcement learning enables cost savings of 1.1% (€120/year) compared to simple rule-based charging. The V2G sell price sensitivity analysis, conducted at 80%, 100%, and 120% of the wholesale purchase price, reveals that V2G discharge arbitrage produces zero revenue at wholesale price levels. This is a structural finding: the absolute price spread at wholesale levels (€0.01–0.06/kWh) is insufficient to overcome roundtrip efficiency losses, regardless of the sell-to-buy ratio. V2G value creation therefore requires either retail-level tariffs, ancillary service markets, or significantly wider price differentials. The integration of solar power into the charging process is important for both the economic viability of the system and its ability to reduce greenhouse gas emissions. Furthermore, smart charging processes maximize the use of solar-generated electricity. Depending upon factors such as fleet size, tariff structures, local utility limitations, and organizational objectives, the optimal level of intelligence required may vary significantly.

5.4.1 Closing Statement.

As electric vehicle penetration increases and renewable resource capacity expands, smart charging will become increasingly important in the transition to a more sustainable energy system. Commercial offices offer an ideal setting for testing smart charging technologies because there is regular occupancy and often a source of renewable energy available (rooftop photovoltaic panels). This thesis demonstrates that coordination between EV charging and building load/PV generation can produce substantial economic benefits as well as environmental reductions.

Additional research, field trials, and policy support will likely be necessary before these concepts mature to the point where they can be implemented widely. However, technical feasibility and economic attractiveness for smart office charging have been established. The remaining question is when and how these systems will grow.

References

- [1] P. Richardson, D. Flynn, and A. Keane, “Optimal charging of electric vehicles in low-voltage distribution systems,” *IEEE Transactions on Power Systems*, vol. 27, no. 1, pp. 268–279, 2012. DOI: [10.1109/TPWRS.2011.2158247](https://doi.org/10.1109/TPWRS.2011.2158247) [Online]. Available: <https://researchrepository.ucd.ie/server/api/core/bitstreams/aa4aa60b-0da7-448c-b25c-a1e321d9c380/content>
- [2] D. Qiu, Y. Wang, W. Hua, and G. Strbac, “Reinforcement learning for electric vehicle applications in power systems: A critical review,” *Renewable and Sustainable Energy Reviews*, vol. 173, p. 113 052, 2023. DOI: [10.1016/j.rser.2022.113052](https://doi.org/10.1016/j.rser.2022.113052) [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364032122009339>
- [3] Z. Wan, H. Li, H. He, and D. Prokhorov, “Model-free real-time EV charging scheduling based on deep reinforcement learning,” *IEEE Transactions on Smart Grid*, vol. 10, no. 5, pp. 5246–5257, 2019. DOI: [10.1109/TSG.2018.2879572](https://doi.org/10.1109/TSG.2018.2879572)
- [4] M. Shin, D.-H. Choi, and J. Kim, “Cooperative management for PV/ESS-enabled electric vehicle charging stations: A multiagent deep reinforcement learning approach,” *IEEE Transactions on Industrial Informatics*, vol. 16, no. 5, pp. 3493–3503, 2020. DOI: [10.1109/TII.2019.2944183](https://doi.org/10.1109/TII.2019.2944183)
- [5] A. Chiş, J. Lundén, and V. Koivunen, “Reinforcement learning-based plug-in electric vehicle charging with forecasted price,” *IEEE Transactions on Vehicular Technology*, vol. 66, no. 5, pp. 3674–3684, 2017. DOI: [10.1109/TVT.2016.2603536](https://doi.org/10.1109/TVT.2016.2603536)
- [6] D. Qiu, Y. Ye, D. Papadaskalopoulos, and G. Strbac, “A deep reinforcement learning method for pricing electric vehicles with discrete charging levels,” *IEEE Transactions on Industry Applications*, vol. 56, no. 5, pp. 5901–5912, 2020. DOI: [10.1109/TIA.2020.2984614](https://doi.org/10.1109/TIA.2020.2984614)
- [7] M. Dorokhova, Y. Martinson, C. Ballif, and N. Wyrsh, “Deep reinforcement learning control of electric vehicle charging in the presence of photovoltaic generation,” *Applied Energy*, vol. 301, p. 117 504, 2021. DOI: [10.1016/j.apenergy.2021.117504](https://doi.org/10.1016/j.apenergy.2021.117504)

- [8] F. Tuchnitz, N. Ebell, J. Schlund, and M. Pruckner, “Development and evaluation of a smart charging strategy for an electric vehicle fleet based on reinforcement learning,” *Applied Energy*, vol. 285, p. 116382, 2021. DOI: [10.1016/j.apenergy.2020.116382](https://doi.org/10.1016/j.apenergy.2020.116382)
- [9] S. Lee and D.-H. Choi, “Dynamic pricing and energy management for profit maximization in multiple smart electric vehicle charging stations: A privacy-preserving deep reinforcement learning approach,” *Applied Energy*, vol. 304, p. 117754, 2021. DOI: [10.1016/j.apenergy.2021.117754](https://doi.org/10.1016/j.apenergy.2021.117754)
- [10] H. M. Abdullah, A. Gastli, and L. Ben-Brahim, “Reinforcement learning based EV charging management systems—a review,” *IEEE Access*, vol. 9, pp. 41506–41531, 2021. DOI: [10.1109/ACCESS.2021.3064354](https://doi.org/10.1109/ACCESS.2021.3064354)
- [11] N. Sadeghianpourhamami, J. Deleu, and C. Develder, “Achieving scalable model-free demand response in charging an electric vehicle fleet with reinforcement learning,” in *Proceedings of the Ninth International Conference on Future Energy Systems (e-Energy '18)*, 2018, pp. 241–245. DOI: [10.1145/3208903.3212042](https://doi.org/10.1145/3208903.3212042)
- [12] K. Park and I. Moon, “Multi-agent deep reinforcement learning approach for EV charging scheduling in a smart grid,” *Applied Energy*, vol. 328, p. 120111, 2022. DOI: [10.1016/j.apenergy.2022.120111](https://doi.org/10.1016/j.apenergy.2022.120111)
- [13] *Deepblue 3.0 pro — JAM54S31 390–415/MR module datasheet*, Monocrystalline PERC half-cut cell module, 405 Wp, 20.7% efficiency (STC), JA Solar Technology Co., Ltd., 2023. [Online]. Available: <https://www.jasolar.com/>
- [14] European Commission Joint Research Centre, *Photovoltaic geographical information system (pvgis)*, Accessed: March 2026, 2023. [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/en/
- [15] European Commission Joint Research Centre, “Elmas: European load model for aggregated sectors,” Joint Research Centre, Tech. Rep., 2020, Dataset includes 18 cluster load profiles and per-class energy statistics.
- [16] ENTSO-E, *Transparency platform: Day-ahead prices*, European Network of Transmission System Operators for Electricity. Regulation (EU) No 543/2013, 2018. [Online]. Available: <https://transparency.entsoe.eu/>
- [17] EPEX SPOT SE, *Day-ahead auction market results — France*, European Power Exchange. Day-ahead clearing prices for the French bidding zone, part of the Single Day-Ahead Coupling (SDAC), 2018. [Online]. Available: <https://www.epexspot.com/en/market-results>
- [18] *Battery-box premium HVM datasheet*, Modular LiFePO₄ battery system, 2.76 kWh per module, ≥ 96% round-trip efficiency (certified by HTW Berlin), BYD Company Ltd., 2024. [Online]. Available: <https://www.bydbatterybox.com/>

- [19] *Tesla model 3 technical specifications*, Tesla, Inc., 2023. [Online]. Available: <https://www.tesla.com/model3/specs>
- [20] “Electric vehicle conductive charging system - part 1: General requirements,” International Electrotechnical Commission, IEC Standard 61851-1, 2017.
- [21] International Renewable Energy Agency, “Renewable power generation costs in 2023,” IRENA, Tech. Rep., 2024, Accessed: March 2026. [Online]. Available: <https://www.irena.org/Publications/2024/Sep/Renewable-Power-Generation-Costs-in-2023>
- [22] U.S. Department of Energy, *Workplace charging for electric vehicles*, Alternative Fuels Data Center, Accessed: March 2026, 2024. [Online]. Available: https://afdc.energy.gov/fuels/electricity_charging_workplace_managing.html
- [23] E. Elgqvist and J. Pohl, “Evaluating utility cost savings for ev charging infrastructure,” National Renewable Energy Laboratory, Tech. Rep. NREL/PR-6A20-75269, 2019, Accessed: March 2026. [Online]. Available: <https://docs.nrel.gov/docs/fy20osti/75269.pdf>
- [24] Fraunhofer ISE, *Energy-charts: Electricity production and spot prices in Europe*, Licensed under CC BY 4.0, data from Bundesnetzagentur / SMARD.de, 2018. [Online]. Available: <https://api.energy-charts.info/>

Chapter A

Code Implementation

This appendix provides key code excerpts from the simulation implementation. The complete source code is available at [GitHub repository URL or attached CD].

A.1 Project Structure

```
1 ChargingAlgorithm/  
2   data/  
3     multi_ev_config.yml           # Charge-only configuration  
4     multi_ev_v2g_config.yml       # V2G configuration  
5     office_single_building_timeseries.csv  
6     irradiance_pvgis.csv         # PVGIS solar data  
7   src/  
8     models/  
9       building.py                 # Building model  
10      electric_vehicle.py          # EV model  
11      grid.py                     # Grid model  
12      multi_ev_charging_system.py  
13      multi_ev_v2g_charging_system.py  
14    simulation/  
15      multi_ev_simulator.py  
16      multi_ev_v2g_simulator.py  
17    utils/  
18      config.py                   # Configuration loader  
19      data_generator.py           # Data loading utilities  
20      visualization.py            # Plotting functions  
21  scripts/  
22    run_simulation.py              # Main entry point  
23  tests/  
24    [unit tests]
```

Listing A.1: Project directory structure

A.2 Core Models

A.2.1 Building Model

```

1 class Building:
2     def __init__(self, energy_consumption_profile, panel_area,
3                   panel_efficiency, peak_solar_irradiance,
4                   battery_capacity, battery_efficiency,
5                   initial_soc, dod, irradiance_profile=None):
6         self.energy_consumption_profile =
7             energy_consumption_profile
8         self.panel_area = panel_area
9         self.panel_efficiency = panel_efficiency
10
11         # Generate solar production profile
12         if irradiance_profile:
13             self.renewable_energy_profile = [
14                 panel_area * panel_efficiency * irradiance / 1000
15                 for irradiance in irradiance_profile
16             ]
17         else:
18             # Fallback: simple sinusoidal pattern
19             self.renewable_energy_profile = self.
20                 _generate_solar_profile()
21
22     def get_net_energy_demand(self, hour):
23         """Calculate net energy demand (positive = needs grid)"""
24         consumption = self.energy_consumption_profile[hour % 24]
25         production = self.renewable_energy_profile[hour % 24]
26         return consumption - production

```

Listing A.2: Building class (simplified)

A.2.2 Electric Vehicle Model

```

1 class ElectricVehicle:
2     def __init__(self, battery_capacity, initial_soc,
3                   max_charge_rate, max_discharge_rate=0,
4                   energy_per_km=0.18, dod=0.9):
5         self.battery_capacity = battery_capacity
6         self.soc = initial_soc

```

```

7         self.max_charge_rate = max_charge_rate
8         self.max_discharge_rate = max_discharge_rate
9         self.min_soc = 1.0 - dod
10
11     def charge(self, energy_kWh):
12         """Charge the EV battery"""
13         max_energy = min(
14             energy_kWh,
15             (1.0 - self.soc) * self.battery_capacity
16         )
17         actual_energy = max_energy * 0.95 # 95% efficiency
18         self.soc += actual_energy / self.battery_capacity
19         return actual_energy
20
21     def discharge(self, energy_kWh):
22         """Discharge the EV battery (V2G)"""
23         max_energy = min(
24             energy_kWh,
25             (self.soc - self.min_soc) * self.battery_capacity
26         )
27         actual_energy = max_energy / 0.95 # Efficiency loss
28         self.soc -= max_energy / self.battery_capacity
29         return actual_energy

```

Listing A.3: ElectricVehicle class (simplified)

A.3 Optimization Algorithms

A.3.1 Simple Rule-Based Algorithm

```

1 def simple_charge_multi(self, hour):
2     """Simple algorithm: Use solar first, then grid"""
3     total_cost = 0
4
5     # Calculate building energy surplus
6     consumption = self.building.energy_consumption_profile[hour]
7     production = self.building.renewable_energy_profile[hour]
8     building_surplus = max(0, production - consumption)
9     building_grid_draw = max(0, consumption - production)
10
11     # Available grid capacity for EVs

```

```

12 total_capacity = self.grid_capacity_per_hour[hour]
13 available_capacity = total_capacity - building_grid_draw
14
15 remaining_surplus = building_surplus
16
17 # Charge each EV in sequence
18 for ev_config in self.evs:
19     ev = ev_config['ev']
20     if hour < ev_config['arrival_time'] or \
21         hour >= ev_config['target_time']:
22         continue
23
24     # Calculate energy needed
25     energy_needed = min(
26         ev.max_charge_rate,
27         (ev_config['desired_soc'] - ev.soc) * ev.
28             battery_capacity
29     )
30
31     # Use surplus first, then grid
32     energy_from_surplus = min(energy_needed, remaining_surplus)
33     energy_from_grid = min(
34         energy_needed - energy_from_surplus,
35         available_capacity
36     )
37
38     total_energy = energy_from_surplus + energy_from_grid
39     if total_energy > 0:
40         ev.charge(total_energy)
41         cost = energy_from_grid * self.grid.get_price(hour)
42         total_cost += cost
43
44     # Update remaining resources
45     remaining_surplus -= energy_from_surplus
46     available_capacity -= energy_from_grid
47
48     return True, -total_cost # Return benefit (negative cost)

```

Listing A.4: Simple charging algorithm

A.3.2 Reinforcement Learning Algorithm

```

1 def rl_charge_multi(self, hour, episodes=2000,
2                     learning_rate=0.1, discount_factor=1.0,
3                     epsilon=0.15):
4     """Multi-agent Q-learning for EV charging"""
5
6     # Discretize state space
7     soc_bins = np.arange(0.2, 1.01, 0.1)
8     actions = ['charge', 'discharge', 'standby'] # V2G
9
10    # Initialize Q-table
11    Q = defaultdict(lambda: np.zeros(len(actions)))
12
13    # Training phase
14    for episode in range(episodes):
15        # Reset EV states
16        ev_states = [initialize_ev_state(cfg) for cfg in self.evs]
17
18        # Simulate episode
19        for h in range(min_hour, max_hour):
20            for ev_id, ev_state in enumerate(ev_states):
21                # Discretize state
22                state = discretize_state(ev_state, h)
23
24                # Epsilon-greedy action selection
25                if np.random.rand() < epsilon:
26                    action_idx = np.random.randint(len(actions))
27                else:
28                    action_idx = np.argmax(Q[state])
29
30                # Execute action and calculate reward
31                reward = execute_action_and_get_reward(
32                    ev_state, actions[action_idx], h
33                )
34
35                # Update Q-value
36                next_state = get_next_state(ev_state, h)
37                best_next_q = np.max(Q[next_state])
38                Q[state][action_idx] += learning_rate * (
39                    reward + discount_factor * best_next_q
40                    - Q[state][action_idx]

```

```

41         )
42
43     # Execution phase: Use learned policy
44     for ev_config in self.evs:
45         state = get_current_state(ev_config, hour)
46         action_idx = np.argmax(Q[state])
47         execute_action(ev_config, actions[action_idx])
48
49     return True, total_benefit

```

Listing A.5: Q-Learning algorithm (simplified)

A.4 Simulation Orchestration

```

1 def run_multi_ev_v2g_simulation(config):
2     """Run complete multi-EV V2G simulation"""
3
4     # Initialize components
5     building = Building(...)
6     grid = Grid(...)
7     evs = [create_random_ev(config) for _ in range(config['n_evs'])]
8     system = MultiEVV2GChargingSystem(building, evs, grid, ...)
9
10    # Define methods to compare
11    methods = {
12        'simple': system.simple_charge_multi,
13        'rl': lambda h: system.rl_charge_multi(h, episodes=15000),
14    }
15
16    # Run simulation
17    results = {name: defaultdict(list) for name in methods}
18
19    for hour in range(24):
20        for name, func in methods.items():
21            # Reset EVs to same initial state
22            reset_ev_states(evs, results[name], hour)
23
24            # Execute algorithm
25            acted, benefit = func(hour)
26

```



```

27         # Record results
28         results[name]['benefit'].append(benefit)
29         results[name]['ev_soc'].append([ev['ev'].soc for ev in
30             evs])
31
32     # Visualize and analyze
33     visualize_results(results, config)
34
35     return results

```

Listing A.6: Main simulation loop

A.5 Configuration Management

A.5.1 Configuration File Format

```

1  # Building Configuration
2  panel_area: 150
3  panel_efficiency: 0.20
4  building_battery_capacity: 80
5  building_battery_efficiency: 0.95
6
7  # EV Fleet Configuration
8  n_ews: 15
9  ev_battery_capacity: 100
10 ev_initial_soc_range: [0.2, 0.3]
11 ev_desired_soc_range: [0.7, 0.8]
12 ev_arrival_window: [6, 9]
13 ev_target_window: [16, 24]
14 max_charge_rate: 22
15 max_discharge_rate: 22
16
17 # Grid Configuration - France EPEX SPOT day-ahead prices
18 # Source: ENTSO-E / Energy-Charts, March 12, 2018 (EUR/kWh)
19 price_profile: [
20     0.0285, 0.0198, 0.0120, 0.0096, # 00-03h
21     0.0190, 0.0420, 0.0447, 0.0483, # 04-07h
22     0.0481, 0.0460, 0.0444, 0.0450, # 08-11h
23     0.0415, 0.0384, 0.0372, 0.0376, # 12-15h
24     0.0422, 0.0502, 0.0620, 0.0552, # 16-19h
25     0.0432, 0.0412, 0.0368, 0.0350 # 20-23h

```

```

26 ]
27 # V2G sell-back at 80% of purchase price
28 sell_price_profile: [
29     0.0228, 0.0158, 0.0096, 0.0076, # 00-03h
30     0.0152, 0.0336, 0.0357, 0.0387, # 04-07h
31     0.0384, 0.0368, 0.0355, 0.0360, # 08-11h
32     0.0332, 0.0307, 0.0298, 0.0301, # 12-15h
33     0.0338, 0.0402, 0.0496, 0.0442, # 16-19h
34     0.0346, 0.0330, 0.0294, 0.0280 # 20-23h
35 ]
36 grid_capacity_per_hour: [80, 80, 80, ...]
37
38 # Algorithm Parameters
39 rl_episodes: 15000
40 rl_lr: 0.15
41 rl_gamma: 1.0
42 rl_epsilon: 0.10

```

Listing A.7: Example configuration file (multi_ev_v2g_config.yml)

A.6 Data Loading Utilities

```

1 def load_irradiance_pvgis(filepath, date_str):
2     """
3     Load PVGIS irradiance data for a specific date.
4
5     Args:
6         filepath: Path to PVGIS CSV file
7         date_str: Date in format 'YYYYMMDD'
8
9     Returns:
10         List of 24 hourly irradiance values (W/m²)
11     """
12     import pandas as pd
13
14     df = pd.read_csv(filepath)
15
16     # Filter for specific date
17     df['date'] = pd.to_datetime(df['time']).dt.strftime('%Y%m%d')
18     df_day = df[df['date'] == date_str]
19

```

```

20 # Extract hourly irradiance
21 irradiance = df_day['G(i)'].values # Global irradiance on
    inclined plane
22
23 # Ensure 24 hours
24 if len(irradiance) < 24:
25     irradiance = np.pad(irradiance, (0, 24 - len(irradiance)))
26
27 return irradiance[:24].tolist()

```

Listing A.8: PVGIS data loader

A.7 Visualization Functions

```

1 def visualize_multi_ev_v2g(results, config, evs, system):
2     """Create comprehensive visualization of simulation results"""
3
4     fig, axes = plt.subplots(4, 1, figsize=(16, 12))
5
6     # Plot 1: EV SoC evolution
7     for method in ['simple', 'rl']:
8         socs = np.array(results[method]['ev_soc'])
9         avg_soc = socs.mean(axis=1)
10        axes[0].plot(results['hours'], avg_soc,
11                    label=method.upper(), linewidth=2)
12
13    axes[0].set_ylabel('Average SoC')
14    axes[0].set_title('EV Fleet State of Charge')
15    axes[0].legend()
16    axes[0].grid(True)
17
18    # Plot 2: Grid usage
19    for method in ['simple', 'rl']:
20        axes[1].plot(results['hours'],
21                    results[method]['grid_usage'],
22                    label=method.upper(), marker='o')
23
24    # Plot grid capacity limit
25    capacity = [config['grid_capacity_per_hour'][h]
26                for h in results['hours']]
27    axes[1].plot(results['hours'], capacity,

```

```

28         'k--', label='Grid Capacity')
29
30     axes[1].set_ylabel('Grid Power (kW)')
31     axes[1].set_title('Grid Power Consumption')
32     axes[1].legend()
33     axes[1].grid(True)
34
35     # Plot 3: Hourly benefit
36     # [Similar plotting code...]
37
38     # Plot 4: Cumulative benefit
39     # [Similar plotting code...]
40
41     plt.tight_layout()
42     plt.savefig('simulation_results_multi_v2g.png', dpi=300)

```

Listing A.9: Results visualization (simplified)

A.8 Usage Instructions

A.8.1 Running Simulations

```

1 # Install dependencies
2 pip install -r requirements.txt
3
4 # Run charge-only simulation
5 python scripts/run_simulation.py # Set SIMULATION_TYPE = "multi_ev
6     "
7
8 # Run V2G simulation
9 python scripts/run_simulation.py # Set SIMULATION_TYPE = "
10     multi_ev_v2g"
11
12 # Run tests
13 pytest tests/

```

Listing A.10: Command line usage

A.8.2 Modifying Parameters

To experiment with different scenarios:

1. Edit configuration file: data/multi_ev_v2g_config.yml
2. Modify parameters (fleet size, prices, capacities, etc.)
3. Run simulation: python scripts/run_simulation.py
4. Results saved to: scripts/simulation_results_*.png

A.9 Dependencies

```
1 numpy>=1.21.0
2 matplotlib>=3.4.0
3 pyyaml>=5.4.0
4 pandas>=1.3.0
5 pytest>=7.0.0          # For testing
```

Listing A.11: requirements.txt

A.10 Testing

```
1 import pytest
2 from src.models.electric_vehicle import ElectricVehicle
3
4 def test_ev_charging():
5     """Test EV charging functionality"""
6     ev = ElectricVehicle(
7         battery_capacity=100,
8         initial_soc=0.3,
9         max_charge_rate=22
10    )
11
12    # Charge for 1 hour at max rate
13    energy = ev.charge(22)
14
15    # Check energy charged (with efficiency)
16    assert abs(energy - 20.9) < 0.1 # 22 * 0.95
17
18    # Check SoC increase
19    expected_soc = 0.3 + 20.9 / 100
20    assert abs(ev.soc - expected_soc) < 0.01
21
```

```

22 def test_grid_capacity_constraint():
23     """Test that grid capacity is respected"""
24     # [Test code...]
25     pass
26
27 def test_target_soc_achievement():
28     """Test that all EVs reach desired SoC"""
29     # [Test code...]
30     pass

```

Listing A.12: Example unit test

A.11 Performance Profiling

```

1 import time
2
3 def benchmark_algorithms():
4     """Compare execution time of different algorithms"""
5
6     algorithms = {
7         'simple': system.simple_charge_multi,
8         'rl': lambda h: system.rl_charge_multi(h, episodes=2000),
9         'mpc': lambda h: system.mpc_charge(h, horizon=6),
10        'pso': lambda h: system.pso_charge(h, n_particles=30),
11        'milp': system.milp_charge,
12    }
13
14    results = {}
15
16    for name, func in algorithms.items():
17        start = time.time()
18
19        for hour in range(24):
20            func(hour)
21
22        elapsed = time.time() - start
23        results[name] = elapsed
24        print(f"{name}: {elapsed:.2f} seconds")
25
26    return results

```

Listing A.13: Algorithm timing comparison

A.12 Code Availability

The complete source code for this thesis is available at:

- **Repository:** [GitHub URL]
- **License:** [Specify license, e.g., MIT, GPL]
- **Documentation:** See `README.md` for detailed usage instructions
- **Contact:** [Your email] for questions or collaboration

A.13 Reproducibility

To reproduce the results in this thesis:

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Download data files (included in repository)
4. Run: `python scripts/run_simulation.py`
5. Results will be generated in `scripts/` folder

Note: RL results may vary slightly due to random initialization. Use `np.random.seed(42)` for deterministic results.

Chapter B

Data Tables and Additional Results

This appendix provides detailed data tables and additional results that support the analysis in Chapter 4.

B.1 Configuration Parameters

B.1.1 Complete Building Parameters

Table B.1: Complete building configuration parameters

Parameter	Value	Unit
<i>Solar PV System (JA Solar DeepBlue 3.0 Pro [13]):</i>		
Reference module	JAM54S31-405/MR	—
Module rated power	405	Wp
Module efficiency (STC)	20.7	%
Cell type	Mono PERC half-cut	—
Module dimensions	1722 × 1134 × 30	mm
Number of modules	74	—
Total panel area	≈ 150	m ²
System efficiency (sim.)	20	%
Peak solar irradiance	1000	W/m ²
Peak power output	30	kW
<i>Building Battery (BYD Battery-Box Premium HVM [18]):</i>		
Chemistry	LiFePO ₄ (LFP)	—
Module capacity	2.76	kWh
Configuration	4 × HVM 19.3 (7 modules each)	—
System capacity	≈ 80	kWh
Round-trip efficiency	≥ 96 (certified HTW Berlin)	%
Simulation efficiency	95	%
Initial SoC	50	%
Depth of discharge limit	80	%
Max charge rate	50	kW
Max discharge rate	50	kW
<i>Energy Profiles:</i>		
Consumption file	office_single_building_timeseries.csv	—
Irradiance file	irradiance_pvgis.csv	—
Simulation date	March 12, 2018	—

B.1.2 Complete EV Fleet Parameters

Table B.2: Complete EV fleet configuration parameters

Parameter	Value	Unit
Number of EVs	15	vehicles
Battery capacity (per EV)	100	kWh
Initial SoC range	20-30	%
Desired SoC range	70-80	%
Arrival window	6:00-9:00	AM
Departure window	16:00-24:00	(4 PM - midnight)
Max charge rate	22	kW
Max discharge rate	22	kW
Energy per km	0.18	kWh/km
Depth of discharge limit	90	%
Minimum SoC	20	%

B.2 Hourly Data Tables

B.2.1 Electricity Price Profile

Table B.3 presents the complete hourly day-ahead electricity prices for the French bidding zone on March 12, 2018, as retrieved from the ENTSO-E Transparency Platform [16] and cross-validated via the Fraunhofer ISE Energy-Charts API [24]. Sell-back prices for the V2G scenario are set at 80% of the purchase price.

Table B.3: Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018

Hour	EUR/MWh	Purchase (€/kWh)	Sell (€/kWh)	Spread (%)
00:00	28.46	0.0285	0.0228	80%
01:00	19.80	0.0198	0.0158	80%
02:00	12.00	0.0120	0.0096	80%
03:00	9.56	0.0096	0.0076	80%
04:00	19.04	0.0190	0.0152	80%
05:00	42.01	0.0420	0.0336	80%
06:00	44.66	0.0447	0.0357	80%
07:00	48.34	0.0483	0.0387	80%
08:00	48.05	0.0481	0.0384	80%
09:00	46.02	0.0460	0.0368	80%
10:00	44.41	0.0444	0.0355	80%
11:00	45.00	0.0450	0.0360	80%
12:00	41.50	0.0415	0.0332	80%
13:00	38.38	0.0384	0.0307	80%
14:00	37.24	0.0372	0.0298	80%
15:00	37.61	0.0376	0.0301	80%
16:00	42.23	0.0422	0.0338	80%
17:00	50.24	0.0502	0.0402	80%
18:00	62.00	0.0620	0.0496	80%
19:00	55.23	0.0552	0.0442	80%
20:00	43.23	0.0432	0.0346	80%
21:00	41.23	0.0412	0.0330	80%
22:00	36.81	0.0368	0.0294	80%
23:00	35.04	0.0350	0.0280	80%
Average	38.67	0.0387	0.0309	80%

B.2.2 Building Energy Profile

Table B.4: Hourly building consumption and solar production

Hour	Consumption (kW)	Solar Production (kW)	Net Demand (kW)
00:00	6.49	0.0	6.49
01:00	6.62	0.0	6.62
02:00	6.39	0.0	6.39
03:00	6.54	0.0	6.54
04:00	6.41	0.0	6.41
05:00	7.47	0.0	7.47
06:00	7.67	0.0	7.67
07:00	8.36	1.25	7.11
08:00	9.74	5.05	4.69
09:00	9.62	5.89	3.73
10:00	9.33	2.09	7.24
11:00	9.68	6.38	3.30
12:00	9.14	5.61	3.53
13:00	9.75	7.24	2.51
14:00	8.98	3.96	5.02
15:00	9.09	14.62	−5.53
16:00	8.84	5.86	2.98
17:00	8.41	0.80	7.61
18:00	8.02	0.0	8.02
19:00	8.21	0.0	8.21
20:00	7.29	0.0	7.29
21:00	6.62	0.0	6.62
22:00	6.81	0.0	6.81
23:00	6.20	0.0	6.20
Total	191.7	58.8	—

Note: Negative net demand indicates surplus solar energy available for EV charging.

B.3 Individual EV Trajectories

B.3.1 EV State of Charge Evolution

Table B.5: Individual EV SoC trajectories (RL method, charge-only)

Hour	EV 1	EV 2	EV 3	...	EV 15	Average
00:00	25%	22%	28%	...	24%	25%
06:00	25%	22%	28%	...	24%	25%
07:00	30%	28%	33%	...	29%	30%
08:00	38%	35%	41%	...	37%	38%
09:00	46%	43%	49%	...	45%	46%
10:00	54%	51%	57%	...	53%	54%
11:00	62%	59%	65%	...	61%	62%
12:00	70%	67%	73%	...	69%	70%
13:00	75%	72%	78%	...	74%	75%
14:00	76%	73%	79%	...	75%	76%
15:00	76%	73%	79%	...	75%	76%
16:00	76%	73%	79%	...	75%	76%

B.4 Annual Simulation Averages (2018)

This section contains the monthly breakdown of the average daily cost throughout the 365 days of the 2018 simulation. It illustrates the seasonal variations in charging costs.

Table B.6: Monthly average daily cost (€) for Charge-Only algorithms in 2018

Month	Simple (€)	RL (€)	RL Savings (%)
January	31.50	31.37	0.4%
February	30.30	30.08	0.7%
March	30.67	30.20	1.5%
April	29.54	29.26	0.9%
May	28.53	28.19	1.2%
June	29.05	28.72	1.1%
July	28.53	27.94	2.1%
August	28.91	28.47	1.5%
September	28.80	28.52	1.0%
October	29.69	29.41	0.9%
November	31.10	30.40	2.3%
December	31.13	31.06	0.2%
Annual Avg	29.79	29.46	1.1%

The V2G daily costs closely mirror the charge-only costs, as the V2G discharge remains zero across all 2018 wholesale price environments simulated.